My House, My Rules: Chinese Aid, Leader Birth Regions, and the Encroachment of Protected Areas

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Abstract

Foreign-funded development projects have long come into conflict with environmental protection efforts, including by encroaching on protected areas (PAs). We investigate the conditions under which countries are more likely to allow foreign aid projects to infringe on PAs. Drawing on the selective enforcement literature, we hypothesize that leaders are more likely to permit incursions into PAs when projects are located in their home region, as local beneficiaries can translate the resulting rents into political support for the incumbent, while the risk of substantial public backlash over environmental violations is comparatively muted. Using geocoded data on 3,675 Chinese infrastructure projects, we find no overall increase in PA encroachments associated with leaders' birth regions. However, the aggregate null masks a positive association in Africa, where projects placed in a leader's birth region are roughly 10 percentage points more likely to encroach on PAs than projects outside the birth region. This association is concentrated in a few countries and in energy, mining, and industry projects. Exploratory analyses suggest that domestic public opinion toward China and the political salience of ethnicity moderate this effect. We find no analogous effect for World Bank projects.

Key Words: Chinese aid, protected areas, leader birth regions, foreign aid, environmental governance

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1 Introduction

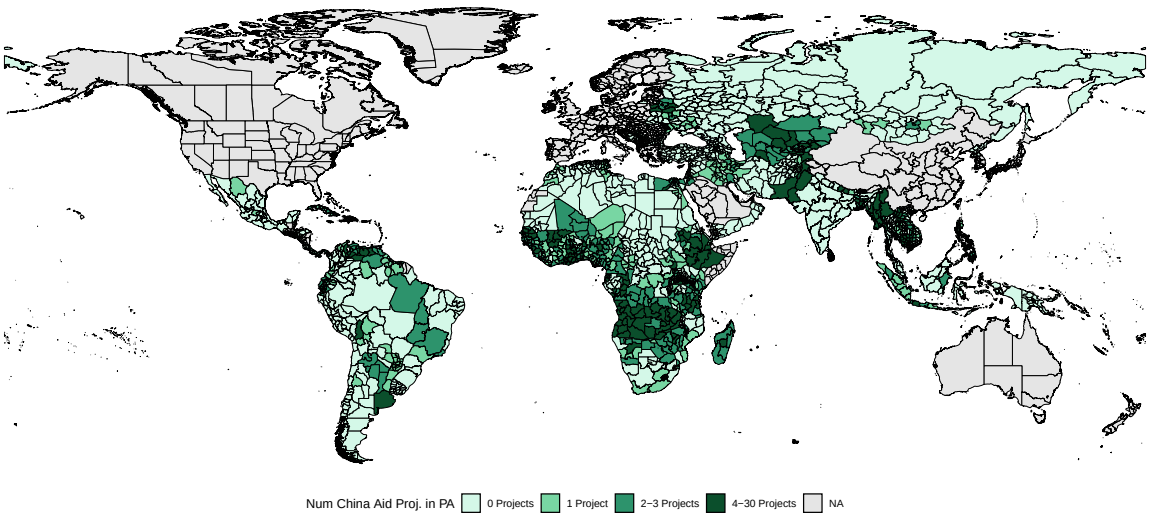
Development assistance projects have long been in tension with environmental protection efforts. For instance, the World Bank has historically been accused of overlooking the environmental impacts of its projects, and even now it funds coal plants across the developing world (Rich 2019). More recently, China – now the world's second largest bilateral aid donor – has received criticism for paying insufficient attention to the environmental consequences of infrastructure projects, with projects in Uganda, Tanzania, the Democratic Republic of Congo, Myanmar, Ecuador, Sri Lanka, and Kenya among the criticized (EJ Atlas 2023; Global Witness 2021; Miller 2019; EJ Atlas 2022b, 2022a; Welz 2019). These projects are notable because their location overlaps with the bounds of protected areas (PAs), geographic areas that place legal limits on commercial activity within their borders and a central tool in international efforts to promote environmental protection (Beacham 2023).

Why are countries tolerating or even promoting projects that potentially violate protected areas? Many explanations would tend to national-level arguments, such as differences in governance quality or state capacity. Yet, when we look at Figure 1, which maps the number of Chinese projects located inside of PAs across ADM1-level subnational units globally, we see significant sub-national variation which is not explained by cross-country differences. Although there is certainly country-level variation, many countries, such as Brazil, exhibit clear variation in PA overlap at the subnational level.[1] This apparent mismatch between law and practice is especially puzzling given that PAs are a popular tool of environmental protection among voters (Michaelsen, Sundström, and Jagers 2025).

In this article, we propose a distributive politics answer to the question of Chinese aid projects in PAs. Drawing on the literatures on selective enforcement (Holland 2015) and home region favoritism (Dreher et al. 2019), we argue that national political leaders selectively permit development projects to infringe on the boundaries of PAs when the proposed project is in the leader's home region. Home region projects are more likely to benefit politically aligned elites, who can then direct some portion of their new profits back to the leader through campaign contributions or other means of support. Because they likely benefit from regional favoritism, home region voters cannot credibly commit to punishing an incumbent for these projects, even if they support environmental protection in principle. In short, allowing home region projects in PAs

1. We show later that this variation is not solely driven by the subnational allocation of protected areas or wealth differences.

Figure 1: Number of Chinese Aid Projects in a PA at ADM1 Level



Note: The figure illustrates the number of Chinese aid projects that impacted PAs within each ADM1 unit between 2000 and 2020. Grey regions indicate countries that have not received any Chinese aid project between 2000 to 2020.

to go forward consolidates elite support with minimal impacts on mass support. This is particularly true when ethnicity is politically salient and China is relatively popular, two scope conditions that indicate the masses are less likely to defect from the incumbent's winning coalition.

Using a sample of 3,675 Chinese infrastructure projects from 2000 to 2020, we estimate the effect of a leader's birth region on the probability that a project is located in a PA. While the aggregate estimate is indistinguishable from zero, we discover a positive association in Africa only, where Chinese aid projects placed in birth regions are significantly more likely to be located within PAs. In contrast, Chinese aid projects placed outside of Africa are no more likely to be within PAs if they are within birth regions. These results are robust to the inclusion of region fixed effects, accounting for spillovers from projects near to PAs, adjustment for post-treatment bias, and alternative specifications with more granular administrative units.[2] We find no such effect for World Bank projects in infrastructure-intensive sectors.

Exploratory analyses of the pooled sample are in line with the scope conditions. We argue that the costs of violating a PA are lower when ethnicity is politically salient and when citizens view China relatively positively. Consistent with these expectations, Chinese aid projects in the leader's birth region are most likely to threaten PAs in country-years where ethnicity is politically salient and public opinion towards China is relatively favorable. These scope conditions may explain why

2. That being said, the results are sensitive to dropping particular countries (Angola and to a lesser extent Ethiopia) as well as excluding particular project sectors.

we observe this relationship between birth region and PA encroachment only in Africa as in Africa, on average, ethnicity is more salient and China is more popular.

This paper makes important contributions to several fields of studies. First, it connects the literature on selective enforcement with scholarship on local impacts of foreign aid. Scholars of comparative politics have convincingly demonstrated that political considerations drive uneven enforcement of legal rules within polities (Holland 2015; Sun 2021; Mattingly 2016), including in the environmental sphere (Harding et al. 2024). Separately, scholarship has shown that foreign assistance is sometimes associated with illegal activity, like corruption (Andersen, Johannesen, and Rijkers 2022; Isaksson and Kotsadam 2018a; Ferry, Hafner-Burton, and Schneider 2020) and violent conflict (Findley et al. 2023). Drawing on theories of selective enforcement, we highlight a case where foreign aid may increase violation of legal rules – in this case, environmental protection laws – not due to issues of agent monitoring or state capacity, but instead with the knowledge and permission of the political leader. We thus add a distributive politics angle to the local effects of foreign aid.

Second, it further investigates the effects of Chinese aid specifically on governance outcomes. Much has been written on how Chinese aid affects economic growth and governance outcomes (Dreher, Fuchs, Hodler, et al. 2021; Dreher, Fuchs, Parks, et al. 2021; Brazys, Elkink, and Kelly 2017; Brazys and Vadlamannati 2021; Isaksson and Kotsadam 2018a, 2018b), yet relatively few studies have touched upon the effects of Chinese aid on environmental outcomes. Studies by Marty et al. (2019) and BenYishay et al. (2016) find that Chinese aid had mixed effects on deforestation. Dreher et al. (2022) record that Chinese aid had a mixed effect on national environmental outcomes. We expand this work on the interaction of Chinese aid with environmental politics by focusing on how Chinese aid conflicts with *de jure* efforts by governments to protect environmentally sensitive areas.

Third, it contributes to research on the efficacy of PAs for environmental conservation. Previous studies on the political economy of PAs suggest that the design and locations of PAs is strongly influenced by political considerations of the government (Mangonnet, Kopas, and Urpelainen 2022; Dawson et al. 2025; Alger 2023; Beacham 2023, 2026). This prior work focuses on how politics at the adoption stage affects PA effectiveness. This study looks at the enforcement stage and how politics potentially undermines PAs, following work from Sanford (2023). Our findings indicate that, under specific conditions, PAs' ability to deter public infrastructure construction may be weakened when the incumbent leader hails from a region benefiting from an infrastructure project.

This speaks to importance of government officials in assuring the effectiveness of PAs (Sundström 2016).

Finally, we advance the study of birth region effects. Previous work theorizes that birth regions are attractive targets for aid in order to curry political favors from leaders (Briggs 2021; Bomprezzi et al. 2024b; Berlin, Desai, and Olofsgård 2023; André, Maarek, and Tapo 2025). Although evidence of birth region effects exists globally (Hodler and Raschky 2014), some scholars have focused specifically on Africa, with the expectation that political dynamics in Africa are more conducive to birth region effects (Dreher et al. 2019). Our evidence is consistent with this expectation. Although we find regional heterogeneity, we also explore scope conditions for which we find suggestive support.

2 Selective Enforcement, Foreign Aid, and Environmental Governance

The decision to enforce the law is shaped by political considerations. While capacity constraints and principal-agent problems are certainly challenges for developing states, politicians strategically decide to enforce certain rules in accordance with their political incentives. For example, South American mayors who court poor voters conduct fewer enforcement operations against unlicensed street vendors, a decision that boosts their support among poor voters (Holland 2015). Depending on the political context, politicians may allow their supporters or swing voters to violate laws that they would enforce on their opponents.

Selective enforcement is a serious challenge to environmental governance, where local elites may oppose protection efforts as it prevents business opportunities. While *de jure* environmental protection has increased significantly as the result of both international and domestic initiatives, *de facto* practices do not always map onto these new rules. In Colombia, for example, Harding et al. (2024) present systematic evidence that campaign contributions reduce regulatory enforcement against illegal forest clearance by local economic elites.

One type of environmental policy vulnerable to selective enforcement is protected areas (PAs). There is strong evidence that PAs can mitigate deforestation (e.g., Andam et al. 2008) and protect biodiversity (e.g., Brodie et al. 2023), but their usage is heavily politicized. Upstream, PA designation is contested, with firms pressuring governments against adopting environmental protections (Alger 2023; Beacham 2023). PAs are concentrated in opposition areas and less

strict when in incumbent strongholds, suggesting distributive politics play a key role (Mangonnet, Kopas, and Urpelainen 2022; Dawson et al. 2025). The map in Figure 1 suggests that enforcement of established PAs is similarly uneven.

Foreign aid can put pressure on recipient governments when these projects clash with existing PAs. Recipient governments have discretion over the allocation of foreign aid and management of its consequences.[3] Recipient governments, while not controlling aid allocation directly, influence where foreign aid is placed in line with their political incentives (Jablonski 2014). Projects may also cause illicit activities in implementation areas, such as corruption (Andersen, Johannesen, and Rijkers 2022; Isaksson and Kotsadam 2018a). These local effects can lead to environmental degradation (Rich 2019). Thus, the location of aid projects (including those with extractive or other environmentally disruptive characteristics) within the boundaries of PAs can represent a threat to environmentally sensitive areas.

Concerns about political capture, selective enforcement, and environmental harm may be particularly salient in the case of Chinese aid. China adopts a more recipient-driven approach to foreign assistance, and evidence suggests this leads to greater sensitivity of its aid to recipient political directives (Dreher et al. 2019; Zeitz 2021; Li 2017; Shea, Reinsberg, and Kern 2025).[4] Consistent with this idea, Chinese aid is also associated with more local corruption (Isaksson and Kotsadam 2018a; Brazys, Elkink, and Kelly 2017) and environmental degradation (Baehr, BenYishay, and Parks 2023), although results on this latter point are mixed (BenYishay et al. 2016). Additional anecdotes are consistent with Chinese tolerance of PA violations in their foreign aid and investment. In East Africa, a Chinese state-owned firm partnered with Tanzania and Uganda to build an oil pipeline through ecologically sensitive areas, including a Ramsar-protected wetland, despite international opposition (EJ Atlas 2023). In Sri Lanka, a Chinese-funded airport near wildlife sanctuaries disregarded environmental guidelines, harming local fauna (EJ Atlas 2022a). Similarly, in Ecuador, a Chinese-owned gold mine polluted waterways serving 1,500 families (EJ Atlas 2022b).

Because recipient national governments are responsible for foreign affairs, they are responsible for stopping Chinese projects that risk violating environmental protection rules. Descriptively, it is clear that enforcement of these rules is uneven. The following section lays out conditions that we

3. We focus on recipient side political considerations. For discussion of donor goals with respect to the environment in particular, see McLean (2015).

4. While we focus on China, our theory could plausibly apply to other donors who allow recipients more control over aid implementation. This includes donors who reject conditionality (Li 2017), those who prioritize political and competitive objectives (Watkins 2021), and those who incorporate borrower voices into institutional governance (Humphrey and Michaelowa 2013).

identify as key for understanding PA enforcement. Drawing on the selective enforcement literature, we theorize that PA enforcement is rooted in recipient's political incentives to enrich politically aligned elites, especially when voters are unlikely to punish them for rule violations.

3 Leader Birth Regions and Selective Enforcement

We consider a decision-theoretic problem for a political leader who is interested in using foreign aid and investment projects to retain political power.[5] In order to achieve this goal, the leader must appeal to both the masses and to other elites.[6] In particular, the leader must demonstrate that a sufficiently large number of each group is better off with the leader in power than they would be under the rule of some challenger. Masses can vote against the leader or take to the streets if unhappy, while elites can withhold campaign contributions and endorse challengers. A leader's main goal is minimize these threats.

One tool leaders use to secure political support is foreign aid. Foreign aid allows politicians to claim credit which allows them to improve their political standing among the public (e.g., Cruz and Schneider 2017; Baldwin and Winters 2023; Dietrich, Mahmud, and Winters 2018; Springman 2022). They can also influence where aid projects are allocated (e.g., strongholds or swing districts) and reap political gains from the distributive effects of aid (Anaxagorou, Efthyvoulou, and Sarantides 2020; Jablonski 2014; Masaki 2018; Hodler and Raschky 2014). Foreign aid is also an opportunity for elites to extract rents from these projects, for instance, by receiving a contract from an aid project (e.g., Andersen, Johannesen, and Rijkers 2022; Bjørnskov 2010). In addition, foreign aid can improve the local business environment and improve local firm profitability (Marchesi, Masi, and Paul 2025). Given these political benefits, there is evidence that politicians pursue aid projects strategically prior to elections (Chen 2025; Kersting and Kilby 2016) and that foreign aid can extend a leader's tenure (e.g., Shea, Reinsberg, and Kern 2025; Licht 2010; Kono and Montinola 2009).

Chinese aid during our study period is particularly vulnerable to be captured by leaders in recipient countries for their political motives. In general, recipients are better able to shape Chinese aid in their favor as Chinese aid projects have fewer strings attached (Dreher et al. 2022).

5. Because the interaction we discuss is international in nature, we think of this veto player as the head of the central government (i.e., a president or prime minister). While regional governments may matter at the margins for these interactions, they likely have less latitude to make selective enforcement decisions in our particular case.

6. Elites encompasses political, social, military, and economic leaders. We focus specifically on the role of firms, who are important actors in both democracies and autocracies.

On the environment in particular, previous official state guidelines only required Chinese firms operating abroad to comply with recipient country regulations; more recent guidelines recommend that transportation projects (but not energy and mining projects) avoid nature reserves (Nedopil, De Boer, and Danting 2022).[7] This orientation towards aid and the environment gives recipients the chance to design domestic policies in line with their preferences (Kaplan 2021) and allocate funding projects in their preferred areas (e.g., the leader's home region) (Dreher et al. 2019; Dreher, Fuchs, Parks, et al. 2021; Bomprezzi et al. 2024b). Furthermore, Chinese aid projects tend to be less transparent, which means they are harder to observe for outside actors reducing accountability pressure for the government (Brown 2025; Cormier 2023). All these features make Chinese aid attractive for leaders to extend their tenure (Shea, Reinsberg, and Kern 2025).

A key challenge for political leaders is that some Chinese aid and infrastructure projects, if implemented, would be placed within the boundaries of PAs. Numerous cases illustrate how Chinese-financed infrastructure has generated controversy due to its location within environmentally sensitive areas. For example, in the Democratic Republic of the Congo (DRC), the government proposed the construction of the USD 500 million Sombwe hydropower dam in collaboration with PowerChina. The project is planned inside Upemba National Park, one of the country's most important protected areas, and has drawn criticism from environmental NGOs as well as international actors. The European Union's ambassador to the DRC publicly questioned the legality of constructing the dam within a national park (Global Witness 2021). A similar case is the Olkaria geothermal power complex in Kenya, which has received Chinese financing.[8] The plants are located within Hell's Gate National Park and have raised concerns about noise pollution and gas emissions that may threaten local wildlife (Welz 2019).

These cases illustrate how economic development objectives can directly clash with conservation commitments. At the same time, important differences exist across projects. Some infrastructure projects, such as the Sombwe Dam, are not inherently location-bound and could in principle be relocated outside a protected area. In such instances, siting the project within a PA reflects a choice rather than a technical necessity. Other projects, however, are more geographically constrained. For example, geothermal power plants like the Olkaria complex must be built where underground heat resources are accessible, limiting the scope for relocation. Nevertheless, even in these cases, developers may regard locations within PAs as economically optimal due to lower land

7. This shift likely indicates increasing Chinese reputational concerns with their projects overlapping with PAs. Assuming firms heeded the recommendations, it is unlikely that the shift coincides with our study period.

8. See https://china.aiddata.org/projects/58350/ (last accessed February 20, 2026).

acquisition costs, proximity to complementary infrastructure, or resource accessibility. Ultimately, the political leader is a veto player: he or she needs to decide whether to enforce environmental protection rules against a proposed project or to allow it to move forward. The final determination therefore reflects not only technical feasibility but also political incentives.

In cases where an aid project intrudes upon environmentally sensitive areas, like a PA, leaders have to deal with competing political pressures. On the one hand, economic elites are likely to favor projects that encroach upon PAs. Development within protected areas can create lucrative opportunities for firms seeking to maximize profits and may involve relaxing or circumventing environmental regulations that would otherwise constrain commercial activity. This logic aligns with research on the political economy of PA placement. Studies show that strong commercial interests in a region shape both the likelihood that a PA is designated (Beacham 2023) and the strictness of its protection (Alger 2023). Moreover, Mangonnet, Kopas, and Urpelainen (2022) demonstrate that in Brazil, protected areas have sometimes been strategically placed in opposition strongholds, potentially to undermine local economic growth.

On the other hand, citizens on average appear more supportive of conservation and may oppose violations of PAs. Public opinion research consistently finds broad backing for protected areas (e.g., Newmark et al. 1993; Michaelsen, Sundström, and Jagers 2025). A large-scale survey conducted in eight countries shows that more than 80% of respondents support the "30-by-30" target adopted in the 2022 Global Biodiversity Framework, which aims to protect 30% of land and water areas by 2030 (Michaelsen, Sundström, and Jagers 2025). Although economic investments could, in principle, offset environmental concerns, evidence suggests that citizens are unwilling to trade environmental protection for economic gains without limits. Individuals tend to reject investments perceived as environmentally irresponsible, even when they promise economic benefits (Nguyen and Malesky 2021; Moehlecke, Fasolin, and Spektor 2025; Beesley and Slaski 2024). Beyond attitudinal support for conservation, violations of protected areas have also been shown to trigger political mobilization, including public protests (Bontempi et al. 2023).

Taken together, this discussion highlights that a leader's decision to permit a project to infringe on PAs is affected by two mechanisms: a rent and a public opinion channel. In the context of PAs, these channels create a political dilemma for leaders: satisfying economic elites by allowing development projects in PAs risks alienating environmentally concerned citizens, while upholding conservation commitments may frustrate powerful commercial interests. Leaders therefore face a strategic choice in balancing these competing constituencies.

How do leaders manage this tradeoff? The political geography of the situation is key. We argue that leaders are more likely to permit projects within PAs when this project is benefiting their home region. The placement in a birth region affects the decision through both the rent as well as the public opinion mechanism. On the public opinion side, we argue that it reduces the costs of backlash. Core voter regions, such as the leader's birth region, are less likely to punish leaders for illegal behavior or under-performance (Carlson 2015; Posner 2005; Zovighian 2026). In addition, these voters also face a credible commitment problem in punishing the incumbent by supporting a challenger, as they fear doing even worse under a challenger's regime with a different power base (e.g., Kasara 2007). Thus, voters in these regions may be more forgiving of selective enforcement of PAs than voters in more competitive districts.

Birth regions also relate to the rents leaders can extract. Selective enforcement in favor of projects benefiting the birth region may bring higher private rents. Elites in the birth region will channel part of the economic gains from an incoming aid project to an incumbent as they have private interests in assuring the leader stays in power and they keep benefiting from new projects (Boubakri, Cosset, and Saffar 2012, 2008; Tawiah, Zakari, and Wang 2022). The main empirical prediction from this argument is the following hypothesis.

Hypothesis 1 (H1) *Leaders are more likely to permit a Chinese project within a PA if the project is in their home region, compared to a project not in their home region.*

3.1 Scope Conditions

Our argument is likely to hold under certain conditions. First, placing an aid project in a leader's home region is most likely to boost the benefits of private rents and reduce the public opinion costs of violating PAs when ethnicity is politically salient.[9] Work on ethnic voting highlights that in societies with high ethnic salience, vote choice tends to be less influenced by a politician's performance, but by the voter's ethnic identity (e.g., Carlson 2015; Ferree, Gibson, and Long 2021; Posner 2005). In these economies, firms likely also have a greater stake in whether their co-ethnic patron is in power. Tawiah, Zakari, and Wang (2022) find that in Nigeria, firms affiliated with the opposition party do not face negative consequences for firm performance if their CEO and the President share ethnic ties.

9. An alternative mechanism in societies without strong ethnic politics that can lead to similar results is affective polarization. While affective polarization is still relatively low in developing countries compared to developed countries, it is increasing (Larreguy and Tiburcio 2026). Diving into this mechanism is outside the scope of this paper, but it is worthwhile to explore how it compares to ethnic salience.

The salience of ethnicity in the context of PAs is highlighted by Dawson et al. (2025). They show that, across Africa, PAs are less likely to be designated in regions inhabited by ethnic groups represented in the national government. Moreover, when such regions do receive a PA, these areas are more likely to permit tourism and other economic activities, suggesting weaker or more flexible forms of protection.

Based on this discussion we expect that in countries with low ethnic salience, voters are more likely to deviate to opposition candidates when they see an encroachment on a PA. Similarly, economic elites may be more likely to fund opposition candidates to avoid negative publicity. These accountability costs maybe be less severe in countries with high ethnic saliency. As a result, we expect the birth region effect to be largest in these countries.

Scope Condition 1 *Leaders are more likely to permit a Chinese project within a PA if the project is in their home region, compared to a project not in their home region, if ethnicity is politically salient in their country.*

Second, we expect that the public opinion mechanism to operate through citizens' views of the donor. Voters are more likely to punish a PA-infringing leader if they hold a negative opinion of the donor (in our case, China). A substantial body of work on foreign direct investment (FDI) and foreign aid shows that there is a plausible link between people's views towards a donor/investor and their support for an aid or FDI project (e.g., Blackman et al. 2025; Findley, Milner, and Nielson 2017; Jud 2023; Kim et al. 2025; Moehlecke, Fasolin, and Spektor 2025; Singh and Williamson 2022). This link between donor favorability and project support could affect attitudes towards an incumbent. Recall that a politician must convince voters that they are better off than they would be under a challenger. Building on research on corruption and political accountability, there is evidence that people overlook corrupt practices if they expect politicians to deliver benefits they like (e.g., Manzetti and Wilson 2007; Klašnja and Tucker 2013; Winters and Weitz-Shapiro 2013). Based on this logic, we expect that a PA violation with a project from a popular donor may be insufficient to cause home region voters to punish the leader, but if it comes from a particular unpopular donor, voters may be less willing to overlook the potential rule violation.

Anecdotes abound of Chinese projects engendering backlash, including in states where the government cooperates with China and when environmental degradation is a risk. In Myanmar – a longtime Chinese partner – popular protests targeted a set of investments by Chinese state-owned enterprises in dams, mines, and oil pipelines. Among the major grievances, according to Miller (2019), were environmental degradation and the exploitation of natural resources. In response to

mounting public pressure, the Burmese president ultimately canceled the projects. This forms our second scope condition.

Scope Condition 2 *Leaders are more likely to permit a Chinese project within a PA if the project is in their home region, compared to a project not in their home region, if China is relatively popular in their country.*

Another common factor to plausibly explain variation in the effect of PAs on Chinese aid placement is democracy (Beacham 2023). The importance of democracy as a moderator is unclear though as it affects the public opinion and rent mechanism in opposing ways. On the one hand, leaders in autocracies need not be as concerned about backlash from voters; elections, if they exist, are not faithful translations of the popular will and meaningful collective action (i.e., protests) are more difficult for the masses to achieve than voting against an unpopular leader. This would imply that the marginal effect of home region decreases as a regime becomes more democratic because violating PAs becomes more costly. On the other hand, leaders in autocracies may not need to use aid projects to extract private rents. They may use more direct methods – like bribes or preferential access to government contracts – that would be unpopular in democracies. If this effect dominates, the marginal effect of home region may not change or would increase in democracies. We remain agnostic on the importance of democracy as a moderator, but include this analysis given the importance of domestic institutions to our theory.

4 Research Design

4.1 Sample

Because our theory operates at the project level, the unit of analysis is the individual aid project rather than a time-series cross-sectional unit such as an ADM1 region-year, as used in related studies (e.g., Dreher et al. 2019; Berlin, Desai, and Olofsgård 2023). The sample includes all Chinese infrastructure projects between 2000 and 2020 for which we have a recorded commitment year and that are implemented in a country for which we know the leader's birth region. We focus on infrastructure projects as these tend to be the most damaging to PAs due to the associated construction activities. According to the Global Chinese Development Finance (GCDF) Dataset

(V3) (Custer et al. 2023; Dreher et al. 2022), 3,675 projects meet these criteria.[10] The sample encompasses projects valued at 615.3 billion US dollars, implemented across 118 countries and covering all types of sectors ranging from social infrastructure (e.g., schools) to mining.[11]

4.2 Treatment and Moderators

Our main treatment variable is a binary indicator set to 1 if a project is implemented in a province (ADM1 level) that corresponds to the birth region of the current national leader.[12] To construct this variable, we leverage the Geospatial Global Chinese Development Finance (GeoGCDF) Dataset (V3), which provides exact locations on Chinese aid projects (Goodman et al. 2024). This allows us to identify all ADM1 regions in which a given aid project is implemented. To determine whether the provinces where projects are implemented align with the leader's birth region, we utilize the newly released Political Leaders' Affiliations Dataset (PLAD) (Bomprezzi et al. 2024a).

One complexity in our dataset is that many projects span multiple provinces (see, for example, Figure 2). We made the decision that if a project crosses any province that is also a leader's birth region, we code the project as treated, i.e., the birth region dummy is coded as 1. This coding decision aligns with our theoretical argument, but it is also conservative, as it may classify projects as treated even when they have only marginal relevance to the leader's birth region (e.g., a railway without a stop in that region).

To evaluate our first scope condition, we interact this home region dummy with country-level variation on the level of ethnic political voting. To measure ethnic voting, we use data compiled by Juon (2024), which combines and standardizes survey questions on support for ethnically based political parties among 900,000 respondents in 132 countries. We aggregate this data at the country-year level, using the share of respondents in a given year who express a preference for ethnic voting as our moderating variable. For some country-years, survey data is unavailable. In these cases, we project the most recent available observation forward.[13]

For our second scope condition, we interact a country's public opinion towards China with

10. Overall, the dataset includes 9,404 projects. Of these, 5,444 were excluded because they are not classified as infrastructure projects. The remaining projects were excluded either due to missing country codes or incomplete commitment year information, which represent a minority of cases.

11. 34% of projects are social development projects, 20% transport projects, 19% energy projects, 12% mining projects, 9% communication and sanitation projects, 5% various industry projects, and 1% are multi-sector projects.

12. As a robustness test, we also code the birth region at the ADM2 level (see C.4).The results in Table C.5 are even clearer than in the ADM1 specification.

13. We acknowledge that this approach relies on the strong assumption that ethnic voting patterns remain unchanged since the last available survey. However, given that ethnic voting is often a deeply entrenched and stable feature in certain societies, we believe this assumption is reasonable.

the home region dummy. Measuring public opinion toward China on a global and temporal scale is challenging. The most comprehensive available dataset is the one compiled by Chubb (2024), which aggregates different surveys on public sentiment toward China.[14] We focus specifically on polls that measure individuals' favorability toward China, using net favorability as our key metric. In country-years with multiple surveys, we calculate the average net favorability across polls, weighting each poll by its sample size.

However, survey data is sparse for many countries, often available only at intervals of three to four years. Given this substantial missingness, assuming that public opinion remains constant between survey years is not reasonable. To address this, we imputed the missing values using Amelia II, a software designed for multiple imputation of time-series cross-sectional data (Honaker, King, and Blackwell 2011). For the imputation, we constructed a country-year panel dataset including all countries with at least one survey observation between 2000 and 2020. We then matched this data with a set of covariates likely to influence public opinion toward China.[15] To ensure robustness, we ran ten imputation models and aggregated the results by calculating the mean imputed public opinion per country-year, aligning it with our project-level Chinese aid dataset.

4.3 Dependent Variable

Our dependent variable measuring PA overlap is called *In PA* and coded as 1 if a project intersects at any point with a PA.[16] As we cannot directly measure whether a project violates a PA since this would require a case-by-case legal analysis, we opt for this more conservative approach where overlap with a PA constitutes a risk for PA violation. Analysis by PA type will give further insights into the violation risk.

To construct our dependent variable of PA overlap, we match geo-location data on the presence of PAs with the locations of Chinese aid projects. We retrieve the geo-locations of all PAs from the World Database on PAs (UNEP-WCMC 2019).[17] We then estimate whether a Chinese project intersects with the boundaries of a PA. For example, Figure 2 illustrates the Kenya SGR project, where the GeoGCDF dataset details the exact route of the project, depicted in red. This enables us to accurately match Chinese aid projects with PAs.

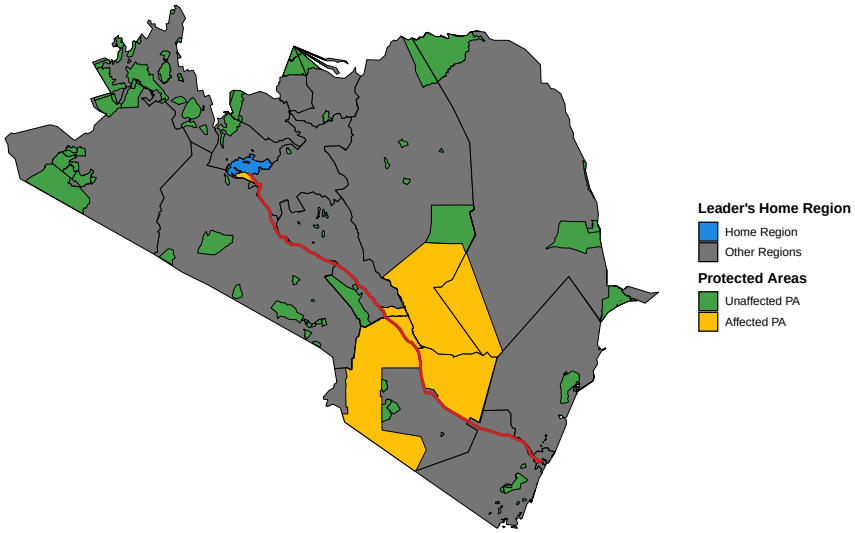
14. These surveys include the Afrobarometer, for example.

15. The covariates included in the imputation are Agreement with China in UN, Trade/GDP (in %), log GDP per capita, oil rents, log population, rents from natural resources, log ad from US, log aid from China, VDEM electoral democracy index, US ally, China ally, communist country, log distance to China, member of AIIB.

16. This applies to both multi-region and single region projects

17. More details on how processed the protected area data is provided in appendix B.

Figure 2: Kenya Standard Gauge Railway and Violations of PAs



Note: Figure shows a subset of counties in Kenya. The county colored in blue is the birth region of the incumbent leader Uhuru Kenyatta. The red line in the represents the route of the SGR. Yellow areas area PAs affected by the SGR and green areas are unaffected PAs in the map area.

Geographically, Chinese aid projects can have four types of different PA overlaps (see Table 1). The first type comprises projects that are not located within any PA, representing the majority of cases. In fact, 78.3% of Chinese aid projects do not overlap with any PAs. While this suggests that most Chinese projects do not directly threaten PAs, it is concerning that 21.7%, or one-fifth, of all Chinese projects are located within PAs, endangering some of the most sensitive environments globally.[18] This issue is further exacerbated by the fact that the most destructive types of projects – transport and energy – have an even higher share within PAs, with 26.8% of transport projects and 24.9% of energy projects located inside PAs.

The second type represents projects which are located within PAs that are both inside and outside a leader's birth region. The Kenya SGR project, shown in Figure 2, serves as a prime example. At the time the SGR was signed, Kenya's leader was Uhuru Kenyatta, whose birth region is the capital, Nairobi. The SGR project encroaches on a PA within Nairobi and also passes through two PAs in counties outside of Nairobi. In our sample, 4% of all projects overlap with PAs both inside and outside a leader's birth region.

The final two type distinguish between projects that overlap with PAs located exclusively outside or inside a leader's birth region. The least common scenario in our sample is when a

18. This is a conservative estimate, as one could argue that projects in close proximity to PAs also pose a threat. When we apply a 1 km and 5 km buffer around Chinese aid projects, the percentage of projects affecting PAs increases to 27.6% and 42.8%, respectively.

Table 1: Classification of PA violations

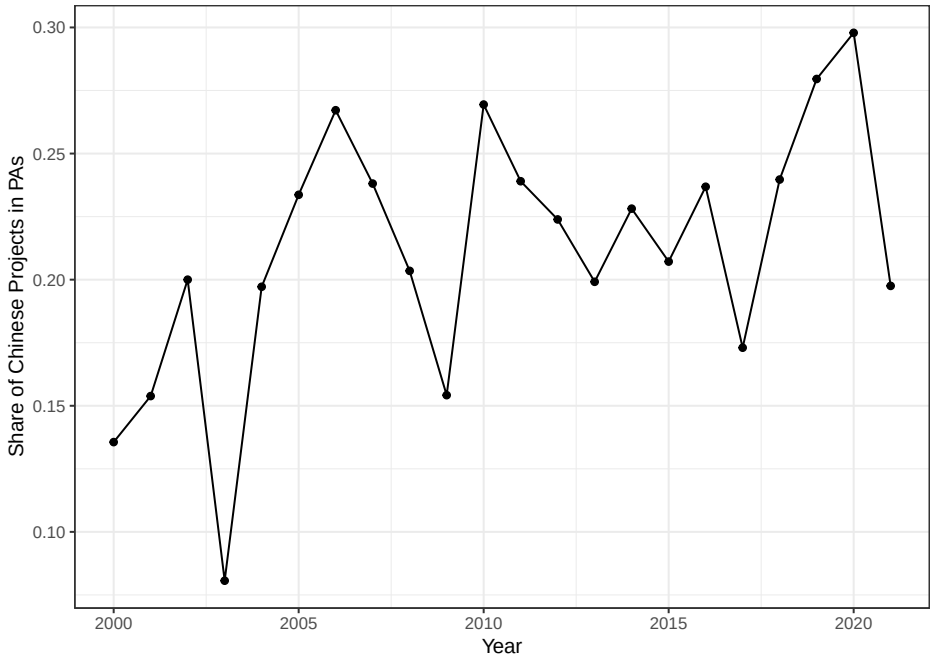
	PA Outside BR = 1	**PA Outside BR = 0**
PA Inside BR = 1	Both (N=148, 4%)	Only Inside (N=28, 0.8%)
PA Inside BR = 0	Only Outside (N=621, 16.9%)	In no PA (N=2878, 78.3%)

Note: PA stands for PA and BR is the abbreviation for birth region. Inside/outside refers to the location of the PA. Of the 891 projects in a birth region, 28 overlap only with PAs inside the birth region, 122 with PAs inside and outside, and 110 only with PAs outside. Percentages in parentheses represent the percentage of cases in our data set that fall under a given category.

project encroaches upon a PA solely within the leader's birth region, accounting for only 0.8% of all Chinese aid projects. Conversely, the second most common scenario involves projects that overlap with PAs only outside the leader's birth region, representing 16.9% of the sample.

Figure 3 shows the temporal development of Chinese aid projects that overlap with PAs. Overall, the share of projects placed within PAs has remained relatively stable, with a slight upward trend compared to the early years of Chinese development assistance. This pattern suggests that there has been no significant structural shift in how Chinese aid projects are sited with respect to environmental protections. Furthermore, the trend does not indicate a clear learning process. There is no evidence that earlier projects were less environmentally conscious and that Chinese actors have become more attentive to environmental concerns over time.

Figure 3: Share of Projects in PAs over Time



4.4 Estimation Strategy

We estimate the relationship between birth region and the location of Chinese aid projects within PAs using a linear probability model as specified in Equation 1 where i denotes an individual project, c the ADM1 regions covered by the project, t the project's start year, and s the project's country. The treatment variable is *BirthRegion*, a binary indicator that equals 1 if a project is implemented in or crosses any ADM1 region (in case of multi-region projects) that is the birth region of the current national leader. This model specification is similar to the approach employed by Dreher et al. (2019).[19]

$$\text{In PA}_{icst} = \beta\text{Birth Region}_{ct} + \delta\mathbf{X}'_c + \lambda\text{Sector}_i + \gamma\text{Num. Regions}_i + \zeta_{st} + \epsilon_{icst} \tag{1}$$

Since the allocation of projects to birth regions is not a random process, our estimation strategy employs a selection-on-observables approach. To control for country-specific confounders, we include country-year fixed effects (ζ_{st}). To address potential sub-national confounders, we incorporate a set of time-invariant control variables at the ADM1 level, denoted by $\mathbf{X}'_c$. These variables include a region's population in 2000, the total number of PAs in 2000, the total number of mines, the total number of oil and gas fields, average nighttime light in 2000, the total area in square kilometers, and whether a project is implemented in a capital region.[20] If a project spans multiple ADM1 regions, we aggregate these variables by taking the mean to create a single observation per project.[21] Finally, we account for project-level characteristics by including sector fixed effects (λ) and controlling for the total number of ADM1 regions a project covers (γ).[22]

Besides the non-random assignment of birth regions, a major challenge for our design is the risk of post-treatment bias. Our sample consists only of projects that have either been realized or have a firm commitment for implementation. However, this sample may be influenced by the presence of birth regions. Dreher et al. (2019) demonstrate that birth regions tend to receive more Chinese aid. Similarly, Briggs (2021) finds in a conjoint experiment with development practitioners that governments are more likely to favor projects in the home region of their leader. Therefore, it is reasonable to assume that by conditioning on project implementation, we may inadvertently introduce post-treatment bias into our estimate.

19. The main difference is that Dreher et al. (2019) use a panel data set and not a project cross-section.
20. For population, nighttime light, and total area, we take the natural logarithm to normalize the distribution.
21. The substantively relevant quantity is the typical condition along a project's footprint, and the mean is the natural summary of that.
22. The latter variable is crucial as multi-region projects just have higher baseline probability to be in a birth region and overlap with a PA.

To address this issue, we employ a strategy proposed by Knox, Lowe, and Mummolo (2020) which allows us to assess the sensitivity of the results to post-treatment bias. Their procedure partitions projects into principal strata defined by whether they would be implemented under each treatment condition, and derives bounds on the treatment effect among implemented projects under a set of transparent assumptions: that unimplemented projects cannot violate PAs, that birth-region status does not make implementation less likely (monotonicity), that projects implemented irrespective of birth region are weakly more likely to overlap with PAs than favoritism projects, and that birth-region status is conditionally ignorable. The result is a range of values rather than a point estimate, allowing us to ask how severe the selection into implementation would have to be to overturn our conclusions. More details are provided in appendix D.

We use cluster-robust standard errors at the recipient country level. Given the likelihood of within-country spillovers in the placement of aid projects, it is reasonable to assume that observations within a country are not independent of one another. In contrast, we are less concerned about spatial spillovers across countries, as our dataset does not include cross-border projects, and project allocation decisions are primarily made by the recipient governments. For these reasons, we do not apply standard error adjustments that account for spatial proximity across countries.

5 Results

5.1 Testing the Main Hypothesis

Table 2 displays the results of our analysis. We also include the results of similar analyses of World Bank projects, which we discuss in greater depth later. Column 1 shows that the presence of a Chinese aid project in a leader's birth region is not significantly related to the likelihood that a project is located in a PA. In fact, the coefficient is nearly zero, suggesting that leaders' decision to respect PA boundaries is unrelated to their birth region. This outcome does not align with hypothesis 1.

Because the pooled estimate is null, we examined whether results vary between projects in African and outside of Africa. We thus interact birth region with a dummy value for Africa. This was motivated by several reasons. First, Africa is more likely to satisfy the scope conditions we outlined in the theory section. African countries express a much higher propensity for ethnic voting

Table 2: Effect of Birth Region on Placement in PA: Pooled Sample and Birth Region × Africa (Chinese and World Bank Aid)

	Pooled				Birth Region × Africa			
Model:	China (i)	World Bank (ii)	China (iii)	World Bank (iv)	China (v)	World Bank (vi)	China (vii)	World Bank (viii)
Variables								
Birth Region	0.006	-0.013	0.090	0.008	-0.116*	-0.031	-0.032	0.020
	(0.041)	(0.042)	(0.073)	(0.051)	(0.048)	(0.046)	(0.122)	(0.049)
Birth Region × Africa					0.225**	0.059	0.234+	-0.032
					(0.069)	(0.081)	(0.137)	(0.099)
Marginal Effect in Africa					0.109*	0.027	0.202**	-0.012
					(0.045)	(0.074)	(0.059)	(0.093)
ADM1 FE	No	No	Yes	Yes	No	No	Yes	Yes
Recipient-Year FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Sector FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Fit statistics								
R^2	0.580	0.764	0.727	0.849	0.586	0.764	0.730	0.849
Observations	3,675	2,117	6,123	10,414	3,675	2,117	6,123	10,414

Note: In columns (v) to (viii), Birth Region is the effect outside Africa, Birth Region × Africa is the difference between the effect in Africa and outside Africa, and the marginal effect in Africa is the sum of the two. Columns (iii), (iv), (vii) and (viii) use project-ADM1 data (one observation per project and ADM1 region, with Birth Region assigned at the project level) and include ADM1 fixed effects in place of region-level covariates, so they control only for the number of ADM1 regions covered by the project; standard errors in these columns are clustered by recipient and project. In the other columns, the models include the following covariates: Number of ADM1 regions covered by project, number of mines, number of oil and gas fields, log area, log population in 2000, log average night time light, number of PAs, and whether the project is in a capital region. All models include recipient-year and project sector fixed effects. The World Bank models include sector dummy variables as a project can cover multiple sectors. Standard errors are clustered by recipient country unless noted otherwise. This estimate rests on 260 projects located in both a birth region and a PA (141 in Africa). Signif. Codes: **: 0.01, *: 0.05, +: 0.1

than non-African countries.[23] Similarly, African countries had an average net favorability toward China of 40.6 percentage points, while outside of Africa, this net favorability was lower, at 31 percentage points. Second, while existing scholarship finds that birth region effects exist globally (Hodler and Raschky 2014), it also finds that effects are strongest in Africa (e.g., Dreher et al. 2019). Finally, Africa is a major target of China's foreign aid spending and broader push for political influence, meaning that Chinese aid to Africa may be particularly sensitive to recipient political interests.

The results in column 5 of Table 2 reveal heterogeneity across continents. Specifically, in African countries, Chinese projects that benefit a leader's home region are 10.9 percentage points more likely to be located in a PA compared to projects placed outside a leader's birth region.[24] Considering that the baseline probability of a Chinese project being located in a PA is 21.7%, this effect represents a 50.2% increase in the likelihood that a project overlaps with a PA. These findings suggest that the aggregate null result is not genuinely indicative of a null but may mask distinct dynamics across different geographic regions.

To gauge robustness of our findings, we create buffer zones around the included Chinese projects. Projects can have spillover effects that may damage PAs even when they are placed outside of them. Thus, selective enforcement may still be relevant in areas in close proximity of PAs. The results in Table C.1 indicate that the null result in the pooled sample remains consistent even when considering projects that are just outside a PA. In Africa, the marginal effect of birth regions on the placement of projects within PAs persists under the 1 km buffer zone specification but is smaller and no longer statistically significant when a 5 km buffer zone is applied. One possible reason is that the 5 km buffer likely includes projects that are not particularly contentious and do not require strong backing from the leader. Outside Africa, the marginal effect is negative under the 1 km buffer (-0.096) and remains negative but is not statistically significant under the 5 km buffer (-0.070). Overall, the results suggest that the projects most affected by the birth region effect are those that actually cross into PAs and needs some form of selective enforcement.

To assess whether our results are driven by particular recipient countries, we conduct a leave-one-out analysis. In the full-sample specification that interacts birth region with the Africa

23. The average ethnic political voting score for projects in Africa is 0.6 and outside of Africa it is 0.19. The ethnic political voting score ranges between 0 and 1 (see Juon 2024).

24. This estimate rests on 260 projects located in both a birth region and a PA (141 in Africa, spread across 33 countries). Of these, 197 (110 in Africa) are in country-years with both birth-region and other projects, which identify the effect under recipient-year fixed effects. The difference between African and non-African is robust to alternative specifications, such as splitting the sample or replacing the Africa dummy with a Sub-Saharan Africa dummy (see appendix C.6).

dummy, the marginal effects of birth region in Africa and outside Africa remain stable across most exclusions, with point estimates and confidence intervals closely tracking the respective full-sample results (Figures C.1 and C.2); the same holds in the split-sample analyses (Figures C.3 and C.4). In the full sample, there are two exceptions: the exclusions of Angola and Ethiopia causes the confidence interval on the marginal effect in Africa to overlap with zero. When we split the sample, the Africa estimate remains significant with the exclusion of Ethiopia, but is insignificant when we exclude Angola. Angola alone accounts for 24 of the 141 African projects that are both in a birth region and in a PA. The African result therefore depends substantially on a single country.

The Africa estimate is sensitive to sectoral composition. Excluding mining, energy, and industry projects reduces the marginal effect of birth region in Africa to 0.075 and renders it statistically insignificant (Table C.3), whereas excluding transport projects leaves it intact (Table C.2). This pattern allows two interpretations. It is consistent with our rent channel, as these sectors generate highly localized rents. However, it may also reflect mechanical overlap, as many of these projects are place-bound and birth regions contain more mines and oil fields (see Tables E.1 and E.2). While our resource controls and the scope condition results make a purely mechanical account less likely, we cannot fully separate the two. We therefore read the Africa result as consistent with, rather than clear evidence of, selective enforcement.

How robust are our estimates from a statistical perspective? We focus on two main sources of potential bias. The first concern is omitted variable bias. Descriptively, projects implemented in a leader's birth region differ systematically from those located elsewhere: they tend to be situated in economically and politically more important areas (see Tables E.1 and E.2). We aim to address this concern by including a comprehensive set of pre-treatment regional covariates. In addition, we run an analysis that expands the dataset to a project-ADM1 dataset, which allows us to include ADM1 fixed effects to control for unobserved regional differences (see details appendix C.5). Column 7 in Table 2 shows a consistent and positive association for African countries with this approach, while the overall relationship remains null (column 3).

Nonetheless, the possibility of unobserved confounders cannot be fully ruled out. The direction of any resulting bias depends on whether governments are systematically more willing to encroach on protected areas in economically valuable regions. Theoretically, arguments can be made for both upward and downward bias. Empirically, however, Tables A.1 to C.8 suggest that our baseline estimates may be somewhat upward biased, as the inclusion of economic controls generally reduces

the magnitude of the birth-region coefficient (though not uniformly across specifications).

A second concern is post-treatment bias as outlined above. To address this issue, we implement the bounding approach proposed by Knox, Lowe, and Mummolo (2020). The results, presented in Figure D.1 in Section D of the supplementary material, indicate that our main estimates are likely conservative.[25] Under plausible assumptions about the extent to which projects are deliberately allocated to benefit the leader's home region, the estimated effect of birth region on PA overlaps increases rather than decreases. Even in the African subsample, where prior research suggests post-treatment bias is most likely, the lower bound of the treatment effect remains above the estimate obtained under the assumption of no post-treatment bias.

Taken together, the overall direction of bias is theoretically ambiguous. Omitted variable bias and post-treatment bias may offset one another or reinforce each other, and either could dominate in magnitude. Importantly, our estimation strategy does not mechanically favor our preferred interpretation.

5.2 Probing the Scope Conditions

We documented an association, concentrated in Africa, between the placement of aid projects in a political leader's birth region the likelihood that these projects are also located within PAs, especially in Africa. In this section, we present exploratory evidence on the role of ethnic voting and public opinion towards China in plausibly moderating this relationship.[26]

As stated in our first scope condition, one source of variation in the costs of PA violation is the salience of ethnicity in a polity's politics. We expect that as ethnic salience increases, a project's connection to a leader's home region will become a stronger predictor of its placement within a PA. To evaluate this claim, we interact the home region dummy with the measure of ethnic political voting.

The results in Figure 4a, which are based on the full project sample, provide suggestive evidence in line with our expectation. A project placed in a birth region has no significant relationship with the placement of a project in a PA in countries with low levels of ethnic voting. However, as ethnic voting becomes more salient, we see that the placement of projects in a leader's home region increase the chance of a PA violation. This is in line with our argument that in more ethnically

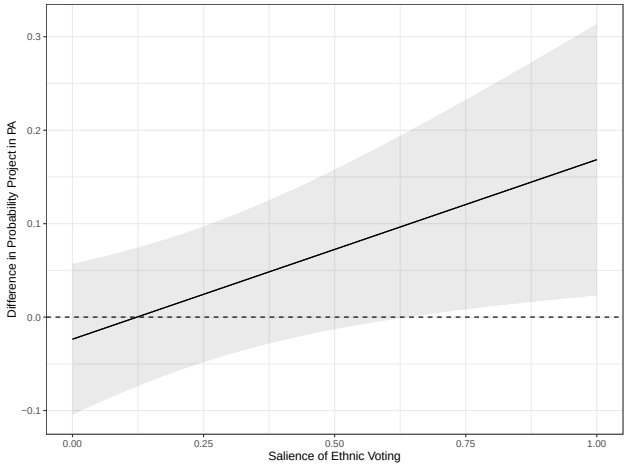
25. To ease implementation and interpretation of the bounding exercise, we use the split sample estimates instead of the marginal effect from the interaction analysis.

26. Again, we rely on the split sample analyses as opposed to the triple interaction in order to ease interpretation.

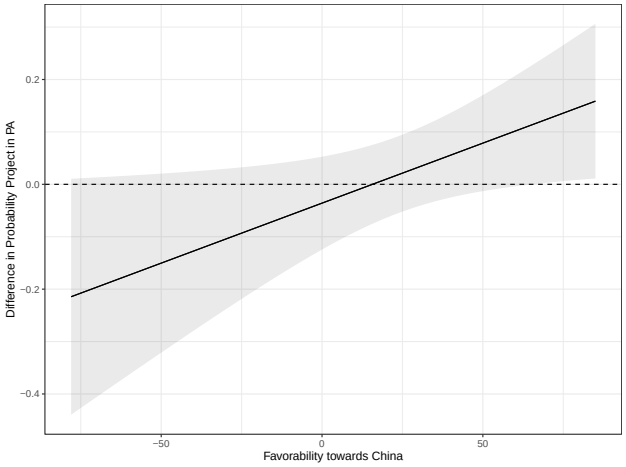
polarized countries, leaders face fewer constraints when dealing with Chinese aid projects.[27]

Figure 4: Moderators of relationship between home region and PA violation

(a) Ethnic political voting



(b) Public opinion towards China



Note: Figure shows marginal effects of birth region on placement in PA conditional the salience of ethnic voting (left panel) and public opinion towards China (right panel) with 95% confidence intervals. Standard errors are clustered by recipient country. The models also include recipient-year and project sector fixed effects.

The second factor that may shape the likelihood of backlash against Chinese projects that infringe on PAs is the favorability of public opinion toward China (see scope condition 2). We predict that leaders may be more reluctant to place aid projects into a PA when sentiment towards China is not favorable even when it benefits the leader's home region.

Based on the full project sample, we find suggestive evidence that local public opinion can influence how a leader's birth region affects the placement of aid projects in PAs (see Figure 4b). As public opinion toward China becomes more favorable, the relationship between birth region and PA violations reverses. At low levels of favorability, projects located in a leader's birth region are less likely to overlap with a PA. However, as net favorability increases, projects in a leader's birth region become more likely to be placed in a PA.[28]

There are three important caveats to these results. Both ethnic salience and public opinion toward China are treated here as conditioning features of the political environment, but each could in principle be shaped by Chinese engagement itself rather than only shaping its consequences. In the case of ethnic voting, we expect that this concern is limited. The measure captures deeply

27. We estimate the analysis separately with the African and non-African samples in C.10. The increase in marginal effect by ethnic salience appears only in the African sample, with ethnic salience making virtually no difference in the non-African sample.

28. We estimate the analysis separately by Africa versus non-Africa in C.9. The change in the marginal effect of birth region is positive across the African versus non-African sample, but the interactions are not statistically significant in either subsample.

entrenched and slow-moving features of a polity's political competition, which are unlikely to respond to the siting of individual aid projects over our sample window. Public opinion toward China is more plausibly endogenous to Chinese activity, and we cannot rule out that project placement feeds back into sentiment.[29] Accordingly, we read them as probing the plausibility of our proposed scope conditions rather than as identified estimates.

A second consideration concerns how we handle the sparsity of the underlying survey data on opinion toward China, which is available only intermittently for most countries. Our main analyses impute the missing country-years through multiple imputation, which necessarily smooths over year-to-year movement in sentiment. We do not claim that the imputed series recovers short-run shifts in public opinion, and the data could not support such high-frequency variation in any case. To ensure that our conclusions are not an artifact of the imputation model or the covariates it draws on, we re-estimate the moderator analyses using an alternative that dispenses with imputation altogether. Following the approach we apply to the ethnic-voting measure, we carry the most recent observed survey value forward to subsequent country-years. The results in Figure C.7 are consistent with those obtained under multiple imputation.

The third question is whether these mechanisms are unique to Chinese aid. We expect that the ethnic salience mechanism should apply to any external finance combining weak conditionality, limited environmental safeguards, and responsiveness to recipient political geography. This is a domestic politics mechanism that alters how much rent leaders can extract (rent mechanism) and how much backlash they need to fear (public opinion mechanism). The donor favorability mechanism is distinct from the ethnic salience mechanism as it relies on people's opinion toward a donor. We predict that this mechanism tends to be likely stronger for bilateral aid than multilateral aid as people general have stronger views towards countries than an international organization. Nonetheless, there is evidence for variation in donor favorability not only in regard to Chinese aid (Blackman et al. 2025; Findley, Milner, and Nielson 2017).

Taken together, these findings suggest that placing aid projects in PAs constitutes a delicate balancing act between the interests of citizens and those of economic elites. The likelihood that a leader permits encroachment on a PA, particularly to promote development in their home region, increases as the political costs of citizen dissatisfaction decline and the benefits of rewarding co-ethnic or in-group constituencies rise. This implies that leaders cannot infringe PAs without

29. To the extent that placement does feed back into sentiment, the resulting bias is likely conservative for scope condition 2. If protected-area violations generate public backlash and depress favorability toward China, this would attenuate the positive association we estimate between favorability and birth-region placement, working against our finding rather than producing it.

constraint. Rather, their decisions are shaped by significant political trade-offs and institutional limits.

An obvious question derived from that result is whether the role of good governance and democracy matters as well. It seems plausible that democratic institutions also represent constraint for leaders. We explore this question by interacting the birth-region indicator with a set of V-Dem governance measures (democracy, clientelism, rule of law, and corruption).

The interactions are not statistically significant (Table E.3). One interpretation is that our scope conditions posit two opposing effects. The public opinion channel predicts a weaker birth-region effect as institutions become more accountable, while the rent channel, if PA violations serve to obfuscate rent-seeking where formal institutions are stronger, predicts the opposite. Because a coarse governance measure loads on both channels at once, their opposing gradients net toward zero. Notably, the moderators that do yield signed effects, ethnic salience and favorability, reinforce each channel, whereas governance does not. However, the results are open to other intepretations, and we do not consider them additional evidence for our theory.

5.3 Comparison to World Bank Projects

Our argument rests on the idea that Chinese aid projects with ties to political leaders' birth regions benefit from selective enforcement of environmental rules. To further support this claim, we examine whether World Bank projects located in birth regions are more likely to be placed within PAs. While prior research shows that the allocation of World Bank projects is also sensitive to a leader's birth region (Berlin, Desai, and Olofsgård 2023), there is evidence that the World Bank more likely follows good governance practices. World Bank projects are generally less susceptible to corruption (Brazys, Elkink, and Kelly 2017; Isaksson and Kotsadam 2018a), owing to its more stringent appraisal procedures, which include environmental safeguards (Buntaine 2015). Accordingly, we expect that due to these institutional differences the placement in birth regions should have a weaker effect on the selective enforcement of environmental rules.

To test this, we use the Geocoded Official Development Aid Dataset (GODAD) (Bomprezzi et al. 2024b) to identify all infrastructure-related World Bank projects between 2000 and 2021 and apply the same spatial matching strategy used for the Chinese aid analysis.[30] This results in a

30. GODAD does not provide a clear indicator for whether a project is infrastructure-related. We follow the approach of Zeitz (2021), who classified World Bank projects as infrastructure-related if they fell into one of the following sectors: water supply and sanitation, transport and storage, communications, energy, agriculture, forestry, fishing, industry, mining, construction, and other social infrastructure.

dataset of 2,123 World Bank projects over that period.

Surprisingly, we find that 29.0% of these projects are located within PAs-a higher share than for Chinese aid projects. However, if a project overlaps with a PA, World Bank projects are less likely to encroach on the most strictly protected categories: only 3.3% intersect with high-stringency PAs, compared to 11% for Chinese-funded projects. This suggests that World Bank projects are more likely to be in line with the goals of a PA and advance projects that are not as harmful to the environment. Moreover, it is important to note that this figure may overestimate the problem in the World Bank case, as we cannot definitively identify which projects involve actual infrastructure components.

In line with our expectations, Table 2 shows that a leader's birth region is not significantly associated with the likelihood that a World Bank project is located within a PA (see columns 2, 4, 6, and 8). The coefficients are substantively small (relative to our China estimates), not consistently signed, and uniformly statistically insignificant at the 95% confidence level. This suggests that, although a higher share of World Bank projects are placed inside PAs overall, this pattern does not appear to be driven by leaders' efforts to selectively enforce rules to benefit their home regions. These findings are consistent with previous research and further strengthen our confidence in the interpretation of the China aid results.[31]

5.4 Sectoral Differences

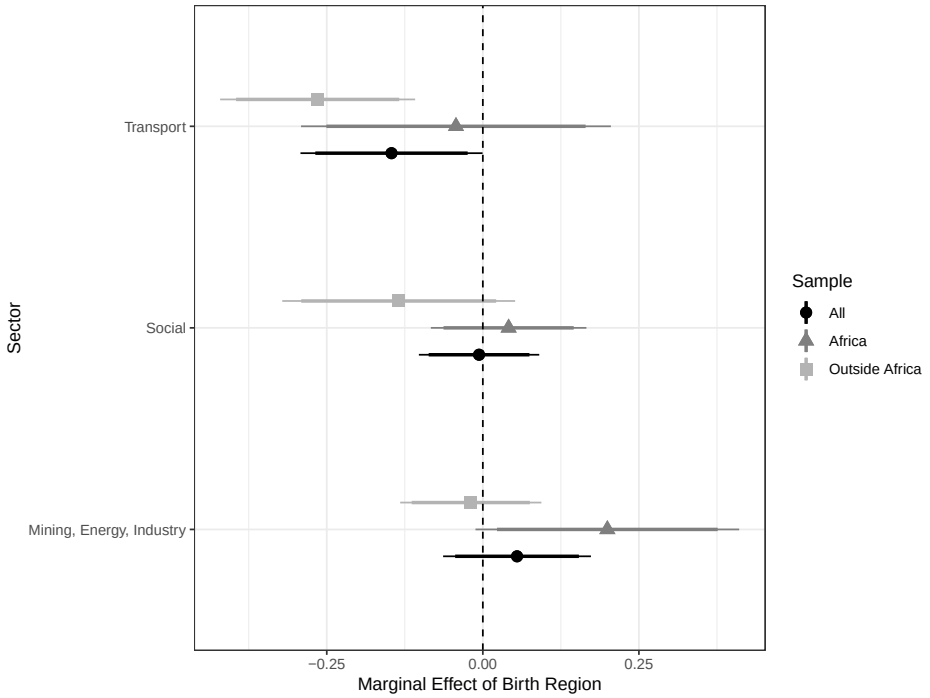
Another issue is that our main results plausibly depend on the sector in which projects are implemented. It is reasonable to expect that not all project sectors allow for the same amount of local rents. For instance, transport projects generate diffuse, system-wide benefits (Bluhm et al. 2025), whereas extractive investments create highly localized rents (Berman et al. 2017; Knutsen et al. 2017). In addition, project implementation in mining and energy may require particular locations with access to resources, whereas projects in other sectors can be moved outside of PAs without great cost. These technological and distributive features make sector-specific heterogeneity in our estimates likely.

Figure 5 shows that this expectation is borne out in the data. In Africa, the marginal effect of birth region is concentrated in projects within the energy, mining, and industry sector (significant

31. A potential threat to this interpretation if the differences stem from project differences. As shown in Table C.11, World Bank and Chinese projects differ quite a bit from each other. To tackle this issue, we combined the two datasets and ran a pooled regression which can account for project differences. The results are consistent with the findings from Table 2. More details and results of this analysis are provided in appendix C.8.

at the 90% level), while the corresponding marginal effects for transport and social projects are statistically indistinguishable from zero.[32] In other words, the aggregate pattern we document does not generalize uniformly across sectors. It is primarily driven by extractive and energy projects in Africa. As mentioned before, while we see this as evidence in line with our selective enforcement argument, it does not fully rule out the possibility that this a result of place-bound projects.

Figure 5: Effect of Birth Region on Project Placement in PAs by Sector



Note: Figure shows the marginal effect of birth region with 95% and 90% confidence intervals. "All" is the effect from the pooled model; "Africa" and "Outside Africa" are the marginal effects from a model that interacts birth region with an Africa dummy (the sum of the birth region coefficient and, for Africa, the interaction coefficient). Standard errors are clustered by recipient country. The models include the following covariates: Number of ADM1 regions covered by project, number of mines, number of oil and gas fields, log area, log population in 2000, log average night time light, and number of PAs. The models also include recipient-year fixed effects.

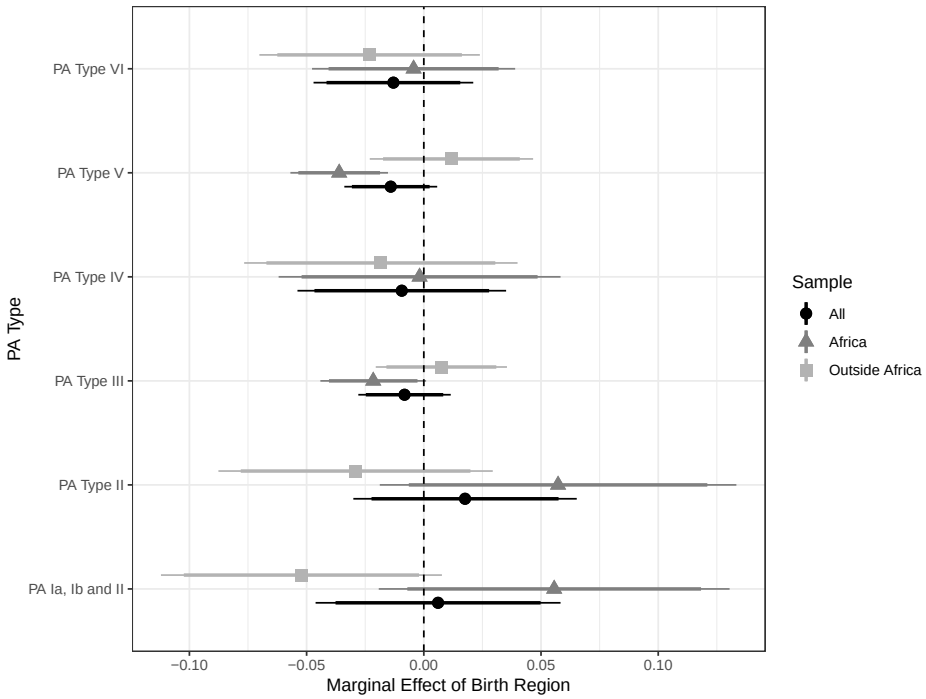
This sectoral profile aligns with our broader interpretation that birth regions matter where local rents are at stake. As mentioned earlier, while we see this as evidence in line with our selective enforcement argument, we cannot fully rule out the possibility that this a result of place-bound projects.

32. There are more than these sectors in the data. We chose to present results for these sectors as they have most projects and make the statistical analysis plausible.

5.5 Types of Protected Areas

If selective enforcement is at play, then we expect that there is also variation in the types of PAs that Chinese aid projects encroach upon. The International Union for Conservation of Nature (IUCN) classifies PAs into seven categories, each differing in the level of permitted human activity. For example, IUCN Category Ia designates "strict nature reserves", where all human activities, except for scientific research, are prohibited. In contrast, IUCN Category VI allows for some form of sustainable resource use. Therefore, we assess whether the birth region dummy is associated with changes in the probability of a project being placed in a PA, depending on the category.

Figure 6: Effect of Birth Region on Project Placement in PAs by Type



Note: Figure shows the marginal effect of birth region with 95% and 90% confidence intervals. "All" is the effect from the pooled model; "Africa" and "Outside Africa" are the marginal effects from a model that interacts birth region with an Africa dummy (the sum of the birth region coefficient and, for Africa, the interaction coefficient). Standard errors are clustered by recipient country. The models include the following covariates: Number of ADM1 regions covered by project, number of mines, number of oil and gas fields, log area, log population in 2000, log average night time light, and number of PAs. The models also include recipient-year and project sector fixed effects. PA types with lower numbers place stricter restrictions on human activities (i.e., PA Types Ia and Ib are the strictest, while Type VI is the least strict).

The results in Figure 6 suggest that the birth region effect is most pronounced in the strictest types of PAs. In African countries, the point estimate marginal effect of birth region on the probability that a project is placed in a PA of IUCN Categories Ia, Ib, and II is positive and substantively largest in stricter PAs, but not statistically significant. This analysis is likely underpowered and at best suggestively consistent with our argument.

6 Alternative Explanations

We argue that the placement of aid projects within PAs is largely shaped by the rent-seeking interests of political leaders and the capacity of citizens to hold their leaders accountable. However, there are also other plausible factors that may deter leaders from placing projects in PAs. One such factor is international economic pressure, especially ties to environmentally-conscious actors (Hawkins and Goodliffe 2023). Garriga and Zhang (2025) find that countries with larger stocks of foreign direct investment (FDI) tend to have more PAs. They argue that these areas enhance a country's international reputation and help protect incumbent foreign firms by raising entry barriers for potential competitors. As such, foreign firms may have an incentive to pressure host governments to respect PAs. Similarly, foreign donors may object to environmentally harmful practices or to the violation of domestic environmental laws. Given these dynamics, we might expect that in countries with greater exposure to international scrutiny–measured by higher levels of FDI and a greater share of GDP derived from official development assistance (ODA)–the influence of birth regions on the placement of Chinese aid projects in PAs should be weaker.[33]

To assess these explanations, we interacted the birth region variable with a country's log FDI stock[34] and the share of ODA relative to the gross national income.[35] If the international pressure explanation applies, we should see a negative and statistically significant interaction coefficient. The results in column 1 and 2 of Table 3, however, do not align with that expectation. The coefficients are very close to zero, positive, and statistically insignificant. This suggests that international pressure is unlikely to influence the extent to which birth regions affect leaders' decisions to place Chinese projects in PAs.

Another possible explanation is that PA encroachment is a function of development. In countries with lower development, leaders may have a stronger motivation to ignore environmental laws and attract Chinese projects for the sake of promoting growth. To test this, we interacted the birth region indicator with two measures of development: the log-transformed GDP per capita at the country level and the log of nighttime light intensity for ADM1 regions covered by a project. Columns 3 and 4 in Table 3 present the results of this test. If development concerns were driving the pattern, we would expect the interaction terms to be negative, indicating a higher likelihood of PA encroachment in less developed birth regions. However, we find no statistically significant

33. We again rely on split sample analysis here to ease interpretation.

34. Following Garriga and Zhang (2025), we do not look at the annual FDI stock, but we take the rolling average mean across four years.

35. FDI data comes from UNCTAD (2024) and the ODA share data was taken from the World Bank (2024).

Table 3: Testing Alternative Explanations

Dependent Variable:	In PA				
Model:	(i)	(ii)	(iii)	(iv)	(v)
Variables					
Birth Region	0.00329	0.00906	-0.03471	0.03227	-0.03962
	(0.18688)	(0.04627)	(0.29231)	(0.09617)	(0.04347)
Num. ADM1 Regions	0.04022*	0.03976*	0.04058*	0.04023**	0.04012**
	(0.01588)	(0.01542)	(0.01588)	(0.01527)	(0.01511)
Num. Mines	0.00139	0.00233	0.00167	0.00219	0.00204
	(0.00211)	(0.00234)	(0.00210)	(0.00220)	(0.00213)
Num. Oil/Gas	0.01188	0.01036	0.01276*	0.01002	0.00933
	(0.00601)	(0.00578)	(0.00501)	(0.00525)	(0.00532)
Log Population	0.03817*	0.03754*	0.03461*	0.03842*	0.04187*
	(0.01765)	(0.01624)	(0.01595)	(0.01595)	(0.01651)
Log Total Area	0.02761	0.03033	0.03097	0.02975	0.02918
	(0.01666)	(0.01553)	(0.01569)	(0.01532)	(0.01531)
Num. PAs	0.00088*	0.00089*	0.00099**	0.00089*	0.00088*
	(0.00039)	(0.00039)	(0.00037)	(0.00037)	(0.00037)
Log Avg. Night Time Lights	-0.04833	-0.04356	-0.04472	-0.03998	-0.03887
	(0.03469)	(0.03219)	(0.03251)	(0.03150)	(0.03086)
ln Capital	0.04183	0.04520	0.04519	0.04506	0.00919
	(0.04019)	(0.03972)	(0.03879)	(0.03845)	(0.03326)
Birth Region × Log FDI Stock	0.00103				
	(0.01965)				
Birth Region × ODA/GNI (in %)		0.00002			
		(0.00369)			
Birth Region × Log GDP/Capita			0.00573		
			(0.03736)		
Birth Region × Log Night Time Ligts				-0.01854	
				(0.04508)	
Birth Region × ln Capital					0.11220
					(0.06392)
Fixed-effects					
Recipient-Year	Yes	Yes	Yes	Yes	Yes
Sector	Yes	Yes	Yes	Yes	Yes
Fit statistics					
R^2	0.574	0.577	0.575	0.580	0.581
Observations	3,430	3,555	3,570	3,675	3,675

Clustered (Recipient) standard-errors in parentheses
*Signif. Codes: **: 0.01, *: 0.05*

interaction effects. While the coefficient for the nighttime lights interaction points in the expected direction, it does not reach statistical significance at the 95% level.

Our findings could also be driven by whether a country's political elites benefits from the project. As most of the political elites are located in the national capital of a country, one may hypothesize that projects going to the capital may be particular popular among these elites. Thus, there will be less political opposition when violating a PA. Column 5 in Table 3 indicates projects located in a birth region and that also benefit the capital region are more likely to be placed inside a PA than a project in a birth region and that does not benefit a capital. This relationship is, however, not statistically significant.

7 Conclusion

Our evidence points to a large but concentrated and context-dependent association between placing a project in a leaders' birth region and the encroachment of PAs. This has several key implications for scholars of international relations and environmental politics. First, they further temper optimism about the efficacy of PAs. Although there is some evidence that PAs can improve local environmental outcomes (e.g., Miranda et al. 2016; Brodie et al. 2023; Andam et al. 2008), extant work points to political interference that prevents the optimal placement of PAs (Alger 2023; Mangonnet, Kopas, and Urpelainen 2022). Our work highlights an additional obstacle to environmental protection: even optimally placed PAs may not be properly enforced if the political incentives for encroaching on them are strong. Nevertheless, PAs create an important focal point for environmental movements to mobilize around. A plausible expectation is that even more aid projects would encroach on environmentally sensitive areas in the absence of PAs.

Second, these findings underscore how political costs can deter leaders from infringing on PAs. Ethnic political voting and public opinion toward China can influence the costs leaders face when enforcing PA boundaries, thereby incentivizing them to minimize such violations. These results suggest that stronger accountability mechanisms can reduce the likelihood that aid projects undermine environmental protection (e.g., Böhmelt and Bernauer 2025).

This project highlights several avenues for future research. First, further research is needed to better understand regional differences in the effects of birth regions. Is our finding specific to PA overlap, or does it extend to the broader distribution of Chinese aid? Investigating whether birth region favoritism influences aid allocation more generally could provide valuable insights into the

political economy of foreign assistance.

Second, survey research on public opinion regarding PAs could offer new insights into why leaders refrain from encroaching on PAs in some areas while doing so in others. This approach could also help disentangle the historical legacies of PAs and their influence on public attitudes toward conservation. As a simplifying assumption, we assume that voters always prefer conservation, but this may not be the case. For example, many conservation policies in Africa are direct legacies of colonialism (Gibson 1999). Perhaps voters only mobilize in favor of PAs designated by their own government and reject externally-imposed PAs. Similarly, identity may also shape the extent to which voters inflict costs for commercial and development activity in PAs. Indigenous rights groups have targeted PAs as vestiges of colonial thinking that displace people who practice response environmental stewardship (Tugendhat et al. 2022).

Third, future research might compare the practices of different donors. While we consider China a most likely case for PA violations and the World Bank a less likely case, we do not investigate why these donors take different approaches. It is possible that bilateral donors prioritize making foreign policy gains and are willing to sacrifice environmental protection. Donors may vary in their interest group environments; perhaps governments in states with large fossil fuel or manufacturing sectors face pressure to fund PA-violating projects, while governments that face large environmentalist interests face pressure to stop such projects.

Fourth, while we analyze the placement of aid projects in PAs, an important next step is to assess their environmental consequences. We assume that placing infrastructure projects, such as energy plants, within PAs has significant environmental impacts, but this remains an open empirical question. Future research should investigate the extent of these impacts, assessing whether aid projects within PAs lead to measurable environmental degradation or if certain mitigation strategies are effective in minimizing harm.

Our findings suggest that international development efforts can interact with domestic political incentives to undermine environmental protection efforts. Crucially, this and other work in environmental politics points to a dire need to take domestic politics seriously in the implementation of environmental protection efforts. If, for example, recent efforts to mobilize billions of dollars in green finance are ignorant of the threat of political interference, they risk falling short of their important goals.

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Supplementary Material "My House, My Rules: Chinese Aid, Leader Birth Regions, and the Violation of Protected Areas"

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A Full Main Results

Table A.1: Effect of Birth Region on PA Placement

Dependent Variable:	In PA							
Model:	(i)	(ii)	(iii)	(iv)	(v)	(vi)	(vii)	(viii)
Variables								
Birth Region	0.170**	0.044	0.039	0.031	0.011	0.010	0.010	0.006
	(0.034)	(0.040)	(0.041)	(0.040)	(0.046)	(0.044)	(0.044)	(0.041)
Num. ADM1 Regions		0.071**	0.067**	0.064**	0.046**	0.041**	0.041**	0.041*
		(0.015)	(0.015)	(0.015)	(0.015)	(0.015)	(0.016)	(0.015)
Num. Mines			0.007*	0.007*	0.003	0.002	0.002	0.002
			(0.003)	(0.003)	(0.002)	(0.002)	(0.002)	(0.002)
Num. Oil/Gas				0.014*	0.009	0.011*	0.010	0.010
				(0.006)	(0.005)	(0.005)	(0.005)	(0.005)
Log Population					0.039*	0.036*	0.045*	0.038*
					(0.016)	(0.016)	(0.017)	(0.016)
Log Total Area					0.043**	0.040**	0.026	0.030
					(0.009)	(0.008)	(0.015)	(0.015)
Num. PAs						0.001*	0.001*	0.001*
						(0.000)	(0.000)	(0.000)
Log Avg. Nighttime Light							-0.037	-0.043
							(0.030)	(0.032)
In Capital								0.045
								(0.039)
Fixed-effects								
Recipient-Year	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Sector	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Fit statistics								
R^2	0.464	0.554	0.557	0.559	0.576	0.579	0.579	0.580
Observations	3,675	3,675	3,675	3,675	3,675	3,675	3,675	3,675

Clustered (Recipient) standard-errors in parentheses
*Signif. Codes: **: 0.01, *: 0.05*

B Coding of Protected Area Data

We construct our measure of PA overlap from the World Database on Protected Areas (UNEP-WCMC 2019). Because national authorities report to the WDPA using heterogeneous national designations and self-assigned IUCN categories, we apply the following harmonization steps.

Record selection. We retain records with status [Designated / Inscribed / Adopted] and exclude those listed as "Proposed", since these do not impose binding restrictions at the time of project siting. We exclude point records for which boundary polygons are unavailable. We restrict attention to terrestrial using the WDPA marine indicator, as our projects are land-based.

Temporal alignment. We use the WDPA designation year to ensure that a project is coded as overlapping a PA only if the area was designated on or before the project's commitment year. This avoids attributing violations to areas protected only after project siting.

Category harmonization. IUCN categories are self-assigned by reporting authorities and applied with varying strictness across countries. We group Categories Ia and Ib, which are functionally similar in the restrictions they impose and individually sparse in our sample. Areas reported as Not Reported, Not Applicable, or Not Assigned account for 35.5% of records overall. These areas are included in our binary "In PA" measure and excluded from the category-specific analysis in Figure 6, as no category can be assigned. Because reporting completeness varies systematically across countries, our country-year fixed effects absorb country-level differences in classification practice.

Residual limitations. Harmonization cannot fully resolve cross-national differences in how strictly a given category is enforced in practice, nor differences in reporting completeness. Our estimates should therefore be read as reflecting de jure protection status rather than de facto enforcement intensity. A distinction consistent with our theoretical focus on whether formal protections constrain siting decisions.

C Robustness Tests

C.1 Buffer Zones

Table C.1: Effect of Birth Region on Placement in PA with Buffer: Pooled Sample and Birth Region × Africa

	In PA (1km)		In PA (5km)	
Model:	Pooled (i)	Birth Region × Africa (ii)	Pooled (iii)	Birth Region × Africa (iv)
Variables				
Birth Region	0.022	-0.096*	0.012	-0.070
	(0.038)	(0.045)	(0.042)	(0.061)
Birth Region × Africa		0.217**		0.152+
		(0.062)		(0.081)
Marginal Effect in Africa		0.121**		0.081
		(0.042)		(0.051)
Recipient-Year FE	Yes	Yes	Yes	Yes
Sector FE	Yes	Yes	Yes	Yes
Controls	Yes	Yes	Yes	Yes
Fit statistics				
R^2	0.583	0.587	0.605	0.607
Observations	3,675	3,675	3,675	3,675

Clustered (Recipient) standard-errors in parentheses
*Signif. Codes: **: 0.01, *: 0.05, +: 0.1*

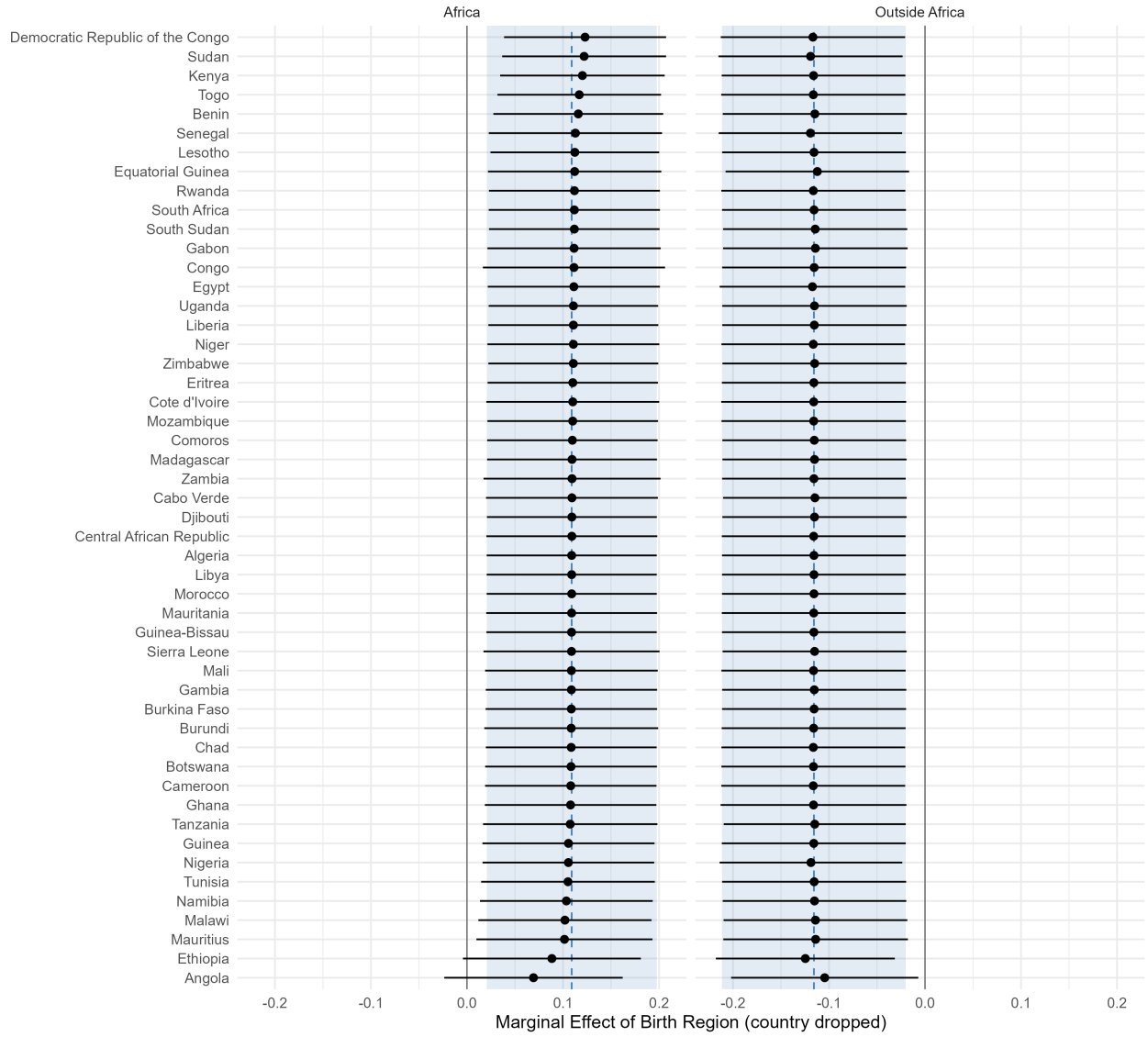
In the Birth Region × Africa columns, Birth Region is the effect outside Africa, Birth Region × Africa is the difference between Africa and outside Africa, and the marginal effect in Africa is the sum of the two. The models include the following covariates: number of ADM1 regions covered by the project, number of mines, number of oil and gas fields, log area, log population in 2000, log average night time light, number of PAs, and whether the project is in a capital region, as well as recipient-year and project sector fixed effects. Standard errors are clustered by recipient country.

C.2 Leave-one-out Analysis

A natural concern with project-level analyses spanning many recipient countries is that the estimated relationship could be shaped disproportionately by one or a small number of countries with unusually large project counts or distinctive characteristics. To assess this, we conduct a leave-one-out (LOO) analysis across three samples: the full sample, the African subsample, and the non-African subsample. For the full sample, we estimate the specification that interacts birth region with an Africa dummy and report the marginal effect of birth region in Africa and outside Africa; for the two subsamples, we estimate the split-sample specification. In each case we re-estimate the model once for each recipient country in turn, excluding that country from the relevant sample. This exercise allows us to characterize the stability of the birth-region coefficient across the exclusion of any single country within each sample, and to identify whether particular cases warrant closer interpretive attention. To keep the full-sample results legible, we split them into two figures by the continent of the excluded country: Figure C.1 excludes African countries and Figure C.2 excludes non-African countries. Both figures show the marginal effect of birth region in Africa and outside Africa.

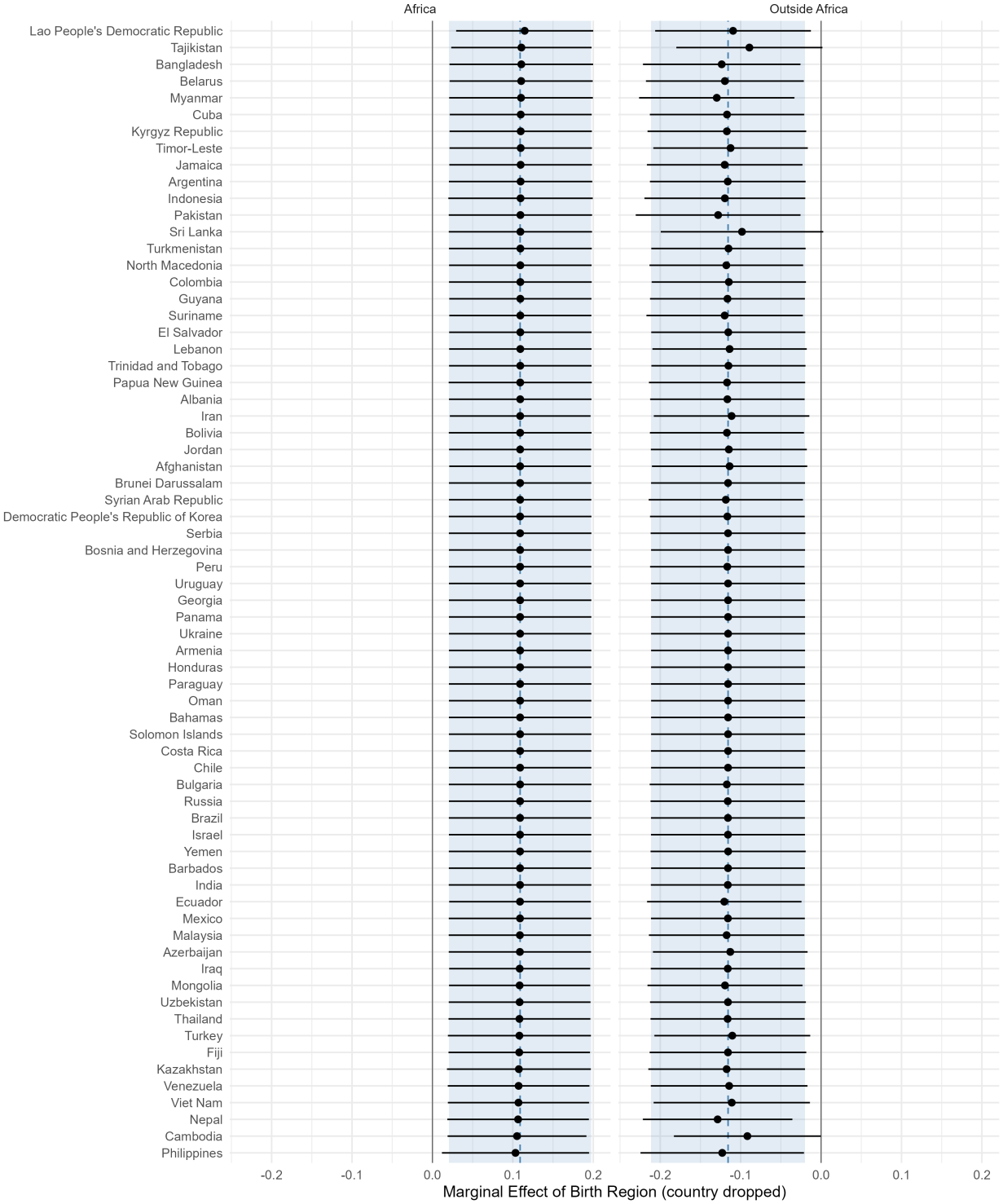
Overall, we find that the birth-region coefficient remains remarkably stable across exclusions in all three samples, with point estimates and confidence intervals largely overlapping their respective full-sample results (Figures C.1 to C.4). The one notable exception arises in the African subsample: when Angola is excluded, the coefficient stays positive, but declines in magnitude and loses statistical significance (see Figures C.1 and C.3). This sensitivity is attributable to two features of the Angolan sub-sample. First, Angola accounts for 230 projects, the largest share of any country in our data, so its exclusion entails a substantial loss of statistical power. Second, Angola's project portfolio is heavily concentrated in mining and energy sectors that account for a disproportionate 11% of all such projects in the Africa sample. This compositional concentration is consistent with our theoretical framework to the extent that these are sectors more malleable to selective enforcement. A sample skewed toward sectors less exposed to this logic will naturally dilute the aggregate estimate once removed. Taken together, these results suggest that Angola's outsized influence reflects its size and sectoral composition rather than a fragility in the underlying birth-region relationship, and is itself informative about the sector-specific channels through which our proposed mechanism operates. No comparable single-country sensitivity is present outside Africa, neither in the full-sample specification nor in the non-African subsample.

Figure C.1: Leave-one-out Analysis: Full Sample, African Countries



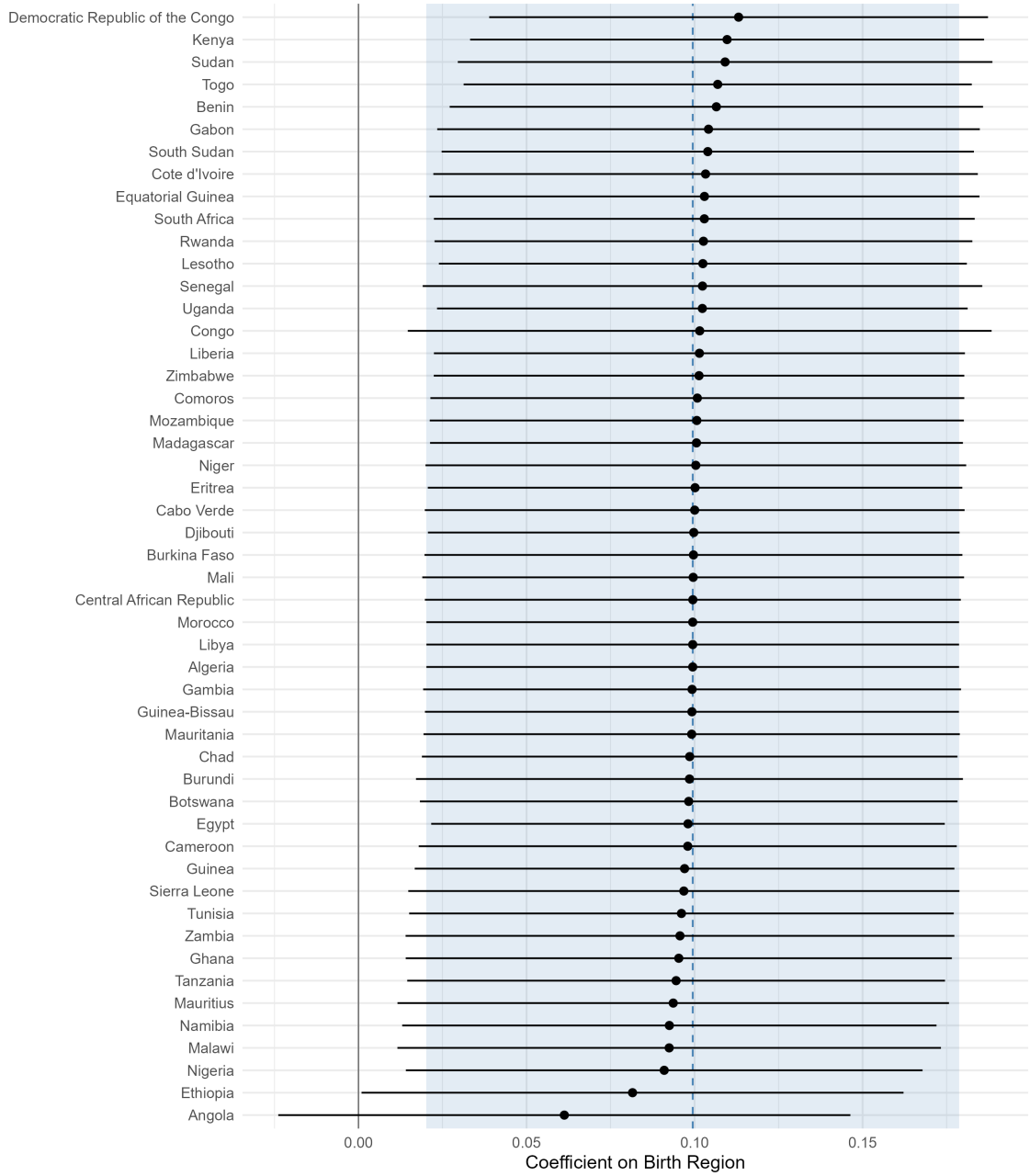
Note: Figure includes point estimates of the marginal effect of birth region in Africa (left panel) and outside Africa (right panel) with 95% confidence intervals, from the model that interacts birth region with an Africa dummy, re-estimated excluding one country at a time. Each row excludes one African country. Dotted line represents the full-sample estimate and shaded area highlights the 95% confidence interval of the full-sample estimate. Y-Axis shows name of excluded country.

Figure C.2: Leave-one-out Analysis: Full Sample, Non-African Countries



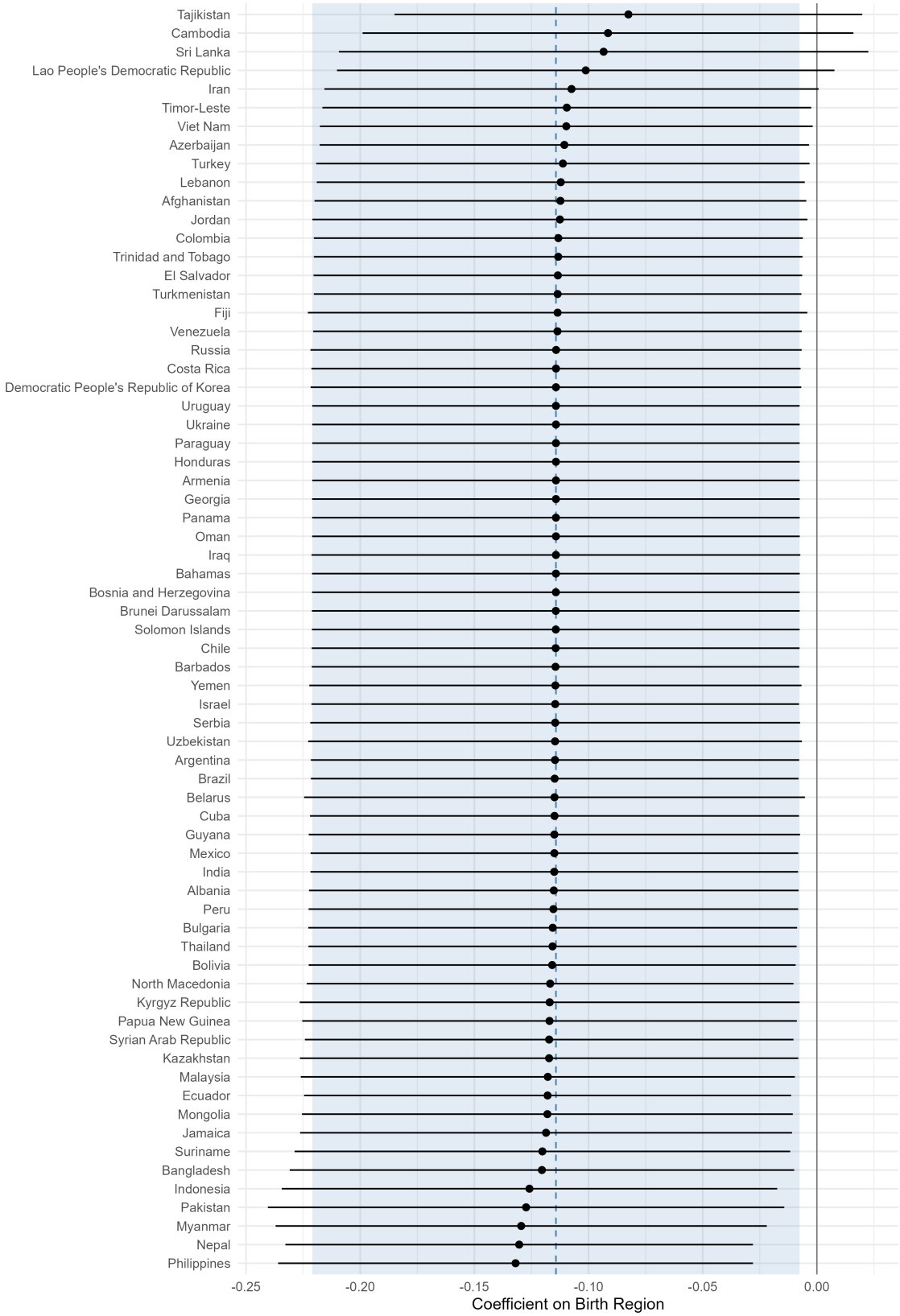
Note: Figure includes point estimates of the marginal effect of birth region in Africa (left panel) and outside Africa (right panel) with 95% confidence intervals, from the model that interacts birth region with an Africa dummy, re-estimated excluding one country at a time. Each row excludes one non-African country. Dotted line represents the full-sample estimate and shaded area highlights the 95% confidence interval of the full-sample estimate. Y-Axis shows name of excluded country.

Figure C.3: Leave-one-out Analysis: Africa Sample



Note: Figure includes point estimates of birth region with 95% confidence intervals. Dotted line represents full sample estimate and shaded area highlights 95% confidence interval of full sample estimate. Y-Axis shows name of excluded country.

Figure C.4: Leave-one-out Analysis: Outside Africa Sample



Note: Figure includes point estimates of birth region with 95% confidence intervals. Dotted line represents full sample estimate and shaded area highlights 95% confidence interval of full sample estimate. Y-Axis shows name of excluded country.

C.3 Sensitivity to Sectors

Table C.2: Main Results Excluding Transport Projects: Pooled Sample and Birth Region × Africa

	Pooled	Birth Region × Africa
Model:	(i)	(ii)
Variables		
Birth Region	0.044	-0.075
	(0.039)	(0.050)
Birth Region × Africa		0.211**
		(0.073)
Marginal Effect in Africa		0.136**
		(0.046)
Recipient-Year FE	Yes	Yes
Sector FE	Yes	Yes
Controls	Yes	Yes
Fit statistics		
R^2	0.625	0.629
Observations	2,933	2,933

Clustered (Recipient) standard-errors in parentheses
*Signif. Codes: **: 0.01, *: 0.05, +: 0.1*

In the Birth Region × Africa columns, Birth Region is the effect outside Africa, Birth Region × Africa is the difference between Africa and outside Africa, and the marginal effect in Africa is the sum of the two. The models include the following covariates: number of ADM1 regions covered by the project, number of mines, number of oil and gas fields, log area, log population in 2000, log average night time light, number of PAs, and whether the project is in a capital region, as well as recipient-year and project sector fixed effects. Standard errors are clustered by recipient country.

Table C.3: Main Results Excluding Mining, Energy, and Industry Projects: Pooled Sample and Birth Region × Africa

	Pooled	Birth Region × Africa
Model:	(i)	(ii)
Variables		
Birth Region	0.000	-0.135*
	(0.042)	(0.059)
Birth Region × Africa		0.211*
		(0.086)
Marginal Effect in Africa		0.075
		(0.050)
Recipient-Year FE	Yes	Yes
Sector FE	Yes	Yes
Controls	Yes	Yes
Fit statistics		
R^2	0.650	0.654
Observations	2,363	2,363

Clustered (Recipient) standard-errors in parentheses
*Signif. Codes: **: 0.01, *: 0.05, +: 0.1*

In the Birth Region × Africa columns, Birth Region is the effect outside Africa, Birth Region × Africa is the difference between Africa and outside Africa, and the marginal effect in Africa is the sum of the two. The models include the following covariates: number of ADM1 regions covered by the project, number of mines, number of oil and gas fields, log area, log population in 2000, log average night time light, number of PAs, and whether the project is in a capital region, as well as recipient-year and project sector fixed effects. Standard errors are clustered by recipient country.

Table C.4: Main Results Excluding Social Projects: Pooled Sample and Birth Region × Africa

	Pooled	Birth Region × Africa
Model:	(i)	(ii)
Variables		
Birth Region	-0.009	-0.118*
	(0.050)	(0.051)
Birth Region × Africa		0.240**
		(0.083)
Marginal Effect in Africa		0.122+
		(0.063)
Recipient-Year FE	Yes	Yes
Sector FE	Yes	Yes
Controls	Yes	Yes
Fit statistics		
R^2	0.629	0.634
Observations	2,411	2,411

Clustered (Recipient) standard-errors in parentheses
*Signif. Codes: **: 0.01, *: 0.05, +: 0.1*

In the Birth Region × Africa columns, Birth Region is the effect outside Africa, Birth Region × Africa is the difference between Africa and outside Africa, and the marginal effect in Africa is the sum of the two. The models include the following covariates: number of ADM1 regions covered by the project, number of mines, number of oil and gas fields, log area, log population in 2000, log average night time light, number of PAs, and whether the project is in a capital region, as well as recipient-year and project sector fixed effects. Standard errors are clustered by recipient country.

C.4 Define Birth-Region at ADM2 Level

Table C.5: Results from Project-Level Analysis at the ADM2 Level: Pooled Sample and Birth Region × Africa

	Pooled	Birth Region × Africa
Model:	(i)	(ii)
Variables		
Birth Region	0.129**	0.084
	(0.039)	(0.054)
Birth Region × Africa		0.075
		(0.080)
Marginal Effect in Africa		0.159**
		(0.054)
Recipient-Year FE	Yes	Yes
Sector FE	Yes	Yes
Controls	Yes	Yes
Fit statistics		
R^2	0.545	0.545
Observations	3,038	3,038

Clustered (Recipient) standard-errors in parentheses
*Signif. Codes: **: 0.01, *: 0.05, +: 0.1*

In the Birth Region × Africa columns, Birth Region is the effect outside Africa, Birth Region × Africa is the difference between Africa and outside Africa, and the marginal effect in Africa is the sum of the two. The models include the following covariates: number of ADM2 regions covered by the project, number of mines, number of oil and gas fields, log area, log population in 2000, log average night time light, number of PAs, and whether the project is in a capital region, as well as recipient-year and project sector fixed effects. Standard errors are clustered by recipient country.

C.5 Accounting for Sub-national Confounders

A concern for empirical analysis is confounding by an omitted variable at the sub-national level. In our main analysis, we aim to account for it by including a set of control variables. A more comprehensive approach commonly used in the literature is the use ADM1 region fixed effects. Yet, with our data structure, this is not feasible. The issue with that is that many of our projects (about 27 percent) span multiple ADM1 regions, meaning that we would either 1) need to drop these observations, which likely represent geographically large and environmentally disruptive projects or 2) arbitrarily choose a region for each observation to be associated with.

A second approach would be to construct a region-year panel data set. However, our concern here is that this approach would involve splitting individual projects into multiple ostensibly independent pieces. Yet, in fact, region-year observations would not be independent from each other due to the presence of projects that span multiple regions. Given that it is not feasible to track the interdependence, we would end up with interdependent observations.

The approach we use seeks to address regional confounding while handling projects that span multiple regions appropriately. We do this by creating a project-ADM1 dataset. Treatment is assigned at the project level, where any project that is at least partially in a birth region is considered treated. Illustratively, imagine a project X that spans regions A (birth region) and B. In our data, there would be two observations that correspond to this project: X-A and X-B. Both observations would be treated, as X is in the birth region A. We view this as the best strategy because selective enforcement that results from a birth region's involvement in a project spills over to the portions of the project that also occur outside of the birth region. We then run our analysis with ADM1 region fixed effects, clustering standard errors both on the region and on the project. In terms of identifying variation, about 97 regions (47 in Africa) have 1) variation in the birth region dummy and 2) at least one Chinese aid project in both treated and untreated periods; the number rises to 422 regions (209 in Africa) when we include regions that host multiregion projects. We also include country-year and sector fixed effects as in our main specification. The results are presented in Table 2 in the main text.

A problem that may arise with the way we defined treatment above is that multi-regional projects receive more weight. While our clustering method means that the standard errors account for the interdependence of observations, our point estimates are disproportionately affected by these projects. To ensure this does not drive our results, we reweight observations such that each project receives equal weight. Accordingly, we assign an observation the weight 1 divided by the number

of ADM1 regions the project overlaps.

The results are presented in Table C.6. The estimates are close to the unweighted estimates (see also Table 2): the marginal effect of birth region in Africa is 0.202 in both specifications and statistically significant at the 1% level, the pooled estimate is 0.112 (0.090 unweighted), and the effect outside Africa remains small and statistically insignificant (0.019 compared to -0.032 unweighted). Multi-regional projects therefore do not drive our results.

Table C.6: Results from Project-ADM1 Level Analysis, Unweighted and with Equal Weight per Project: Pooled Sample and Birth Region × Africa

	Unweighted		Equal Project Weights	
	Pooled	Birth Region × Africa	Pooled	Birth Region × Africa
Model:	(i)	(ii)	(iii)	(iv)
Variables				
Birth Region	0.090	-0.032	0.112*	0.019
	(0.073)	(0.122)	(0.053)	(0.079)
Birth Region × Africa		0.234+		0.183+
		(0.137)		(0.110)
Marginal Effect in Africa		0.202**		0.202**
		(0.059)		(0.066)
ADM1 FE	Yes	Yes	Yes	Yes
Recipient-Year FE	Yes	Yes	Yes	Yes
Sector FE	Yes	Yes	Yes	Yes
Controls	Num. ADM1	Num. ADM1	Num. ADM1	Num. ADM1
Fit statistics				
R^2	0.727	0.730	0.664	0.666
Observations	6,123	6,123	6,123	6,123

Clustered (Recipient & Project) standard-errors in parentheses
*Signif. Codes: **: 0.01, *: 0.05, +: 0.1*

In the Birth Region × Africa columns, Birth Region is the effect outside Africa, Birth Region × Africa is the difference between Africa and outside Africa, and the marginal effect in Africa is the sum of the two. All columns use project-ADM1 data (one observation per project and ADM1 region, with Birth Region assigned at the project level). In the equal project weights columns, observations are weighted by 1 / (number of ADM1 regions the project is in), so that each project carries a total weight of one. Regional covariates are absorbed by the ADM1 fixed effects. Standard errors are clustered by recipient country and project.

C.6 Alternative Estimation of Africa - Outside Africa Sample Split

Table C.7: Effect of Birth Region on PA Placement in Africa

Dependent Variable:	In PA							
Model:	(i)	(ii)	(iii)	(iv)	(v)	(vi)	(vii)	(viii)
Variables								
Birth Region	0.214**	0.096*	0.095*	0.094*	0.096*	0.094*	0.101*	0.099*
	(0.038)	(0.038)	(0.039)	(0.036)	(0.043)	(0.042)	(0.041)	(0.039)
Num. ADM1 Regions		0.099**	0.096**	0.096**	0.074**	0.068**	0.072**	0.072**
		(0.010)	(0.011)	(0.010)	(0.010)	(0.013)	(0.014)	(0.014)
Num. Mines			0.002	0.002	0.000	0.000	0.000	0.000
			(0.002)	(0.002)	(0.003)	(0.003)	(0.003)	(0.003)
Num. Oil/Gas				0.000	0.008	0.010	0.008	0.008
				(0.017)	(0.020)	(0.018)	(0.018)	(0.017)
Log Population					0.030	0.030	0.045*	0.044**
					(0.017)	(0.018)	(0.018)	(0.016)
Log Total Area					0.036**	0.035**	-0.001	0.000
					(0.009)	(0.009)	(0.019)	(0.020)
Num. PAs						0.000	0.000	0.000
						(0.000)	(0.000)	(0.000)
Log Avg. Night Time Light							-0.098*	-0.099*
							(0.044)	(0.045)
In Capital								0.007
								(0.047)
Fixed-effects								
Recipient-Year	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Sector	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Fit statistics								
R^2	0.457	0.591	0.591	0.591	0.603	0.604	0.607	0.607
Observations	1,850	1,850	1,850	1,850	1,850	1,850	1,850	1,850

Clustered (Recipient) standard-errors in parentheses
*Signif. Codes: **: 0.01, *: 0.05*

Table C.8: Effect of Birth Region on PA Placement Outside of Africa

Dependent Variable:	In PA							
Model:	(i)	(ii)	(iii)	(iv)	(v)	(vi)	(vii)	(viii)
Variables								
Birth Region	0.103*	-0.031	-0.039	-0.057	-0.112*	-0.115*	-0.115*	-0.114*
	(0.048)	(0.061)	(0.061)	(0.058)	(0.056)	(0.055)	(0.054)	(0.053)
Num. ADM1 Regions		0.055**	0.053**	0.047**	0.034*	0.031*	0.031*	0.031*
		(0.017)	(0.017)	(0.017)	(0.015)	(0.016)	(0.015)	(0.015)
Num. Mines			0.008	0.006	0.003	0.003	0.003	0.003
			(0.005)	(0.005)	(0.004)	(0.003)	(0.003)	(0.003)
Num. Oil/Gas				0.022**	0.014*	0.014*	0.014*	0.013*
				(0.007)	(0.006)	(0.006)	(0.006)	(0.006)
Log Population					0.048*	0.044	0.044	0.039
					(0.022)	(0.023)	(0.029)	(0.028)
Log Total Area					0.044*	0.041*	0.042	0.045
					(0.017)	(0.016)	(0.026)	(0.026)
Num. PAs						0.001	0.001	0.001
						(0.001)	(0.001)	(0.001)
Log Avg. Nighttime Light							0.002	-0.002
							(0.045)	(0.046)
In Capital								0.037
								(0.040)
Fixed-effects								
Recipient-Year	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Sector	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Fit statistics								
R^2	0.490	0.555	0.558	0.568	0.582	0.583	0.583	0.584
Observations	1,825	1,825	1,825	1,825	1,825	1,825	1,825	1,825

Clustered (Recipient) standard-errors in parentheses
*Signif. Codes: **: 0.01, *: 0.05*

Table C.9: Estimating Difference Between Africa and Outside Africa in Full Sample

Dependent Variable:	In PA	
	All	
Model:	(i)	(ii)
Variables		
Birth Region	-0.116*	-0.107*
	(0.048)	(0.048)
Num. ADM1 Regions	0.042**	0.042**
	(0.015)	(0.015)
Num. Mines	0.002	0.002
	(0.002)	(0.002)
Num. Oil/Gas	0.012*	0.012*
	(0.005)	(0.005)
Log Population	0.050**	0.050**
	(0.016)	(0.016)
Log Total Area	0.026	0.026
	(0.015)	(0.016)
Num. PAs	0.001*	0.001*
	(0.000)	(0.000)
Log Avg. Nighttime Light	-0.049	-0.047
	(0.030)	(0.031)
In Capital	0.025	0.026
	(0.034)	(0.034)
Birth Region × Africa	0.225**	
	(0.069)	
Birth Region × Sub-Saharan Africa		0.214**
		(0.070)
Fixed-effects		
Recipient-Year	Yes	Yes
Sector	Yes	Yes
Fit statistics		
R^2	0.586	0.585
Observations	3,675	3,675

Clustered (Recipient) standard-errors in parentheses
*Signif. Codes: **: 0.01, *: 0.05*

C.7 Difference African to Non-African Projects

Table C.10: Balance table: Comparing African Projects to Non-African Projects

Variable	Mean Africa	Mean Non-Africa	Diff.-in-Means	t Stat.	$p < 0.05$
In PA (Share)	0.22	0.21	0.01	-0.78	No
In Birth Region (Share)	0.23	0.25	-0.01	1.04	No
Number of PAs	22.36	16.86	5.51	-3.41	Yes
Commitment Value (IHS)	17.83	18.61	-0.78	10.24	Yes
Energy Proj.	0.13	0.25	-0.11	8.91	Yes
Capital	0.49	0.32	0.17	-10.49	Yes
Other Industry Proj.	0.06	0.04	0.01	-2.01	Yes
Log Area	9.45	9.62	-0.17	2.31	Yes
Log Night Time Lights	1.70	1.80	-0.10	4.56	Yes
Log Population	14.22	14.74	-0.52	10.35	Yes
Log Population Density	4.81	5.15	-0.34	5.67	Yes
Number of Mines	2.52	1.77	0.75	-3.52	Yes
Mining Proj.	0.05	0.18	-0.13	12.62	Yes
Number Oil Fields	0.38	1.71	-1.32	15.79	Yes
Commu. & Sanitation Proj.	0.12	0.06	0.06	-6.13	Yes
Social Proj.	0.44	0.25	0.20	-12.69	Yes
Transport Proj.	0.19	0.21	-0.02	1.36	No

C.8 Comparison of Chinese Aid and World Bank Aid

Table C.11 compares the characteristics of Chinese and World Bank-financed projects in our sample. Chinese projects tend to be sited in geographically smaller and less densely populated areas, with substantially lower log area, log population density, and number of ADM1 units spanned relative to World Bank projects, and are located in regions with far fewer protected areas, mines, and oil fields on average. Despite being implemented in smaller and more sparsely populated regions, Chinese projects are associated with modestly higher log nighttime lights, log population, and log commitment value, and are somewhat more likely to be sited in a capital region. In terms of sectoral composition, Chinese projects are considerably less concentrated in social, communication and sanitation, and other industry sectors, but are more likely to be energy projects, while mining and transport shares are statistically indistinguishable across the two donors. Overall, the two donors' portfolios differ systematically along both geographic and sectoral dimensions, underscoring the importance of controlling for these characteristics when comparing donor-specific effects.

Table C.11: Balance table: Comparing Chinese Projects to World Bank Projects

Variable	Mean China	Mean World Bank	Diff.-in-Means	t Stat.	$p < 0.05$
Number of PAs	19.63	755.18	-735.55	11.01	Yes
Energy Proj.	0.19	0.11	0.08	-9.59	Yes
Capital	0.41	0.37	0.04	-3.02	Yes
Other Industry Proj.	0.05	0.17	-0.12	16.47	Yes
Log Area	9.53	12.95	-3.42	58.47	Yes
Log Commitment Value	17.54	16.85	0.69	-5.93	Yes
Log Night Time Lights	1.75	1.66	0.09	-5.99	Yes
Log Population	14.48	17.67	-3.19	58.97	Yes
Log Population Density	4.98	4.78	0.20	-4.91	Yes
Number of Mines	2.15	89.11	-86.96	7.28	Yes
Mining Proj.	0.12	0.11	0.00	-0.42	No
Number ADM1 Units	1.90	5.31	-3.41	25.21	Yes
Number Oil Fields	1.04	66.66	-65.62	10.32	Yes
Comm. & Sanitation Proj.	0.09	0.22	-0.13	15.16	Yes
Social Proj.	0.34	0.56	-0.22	18.86	Yes
Transport Proj.	0.20	0.20	0.00	-0.49	No

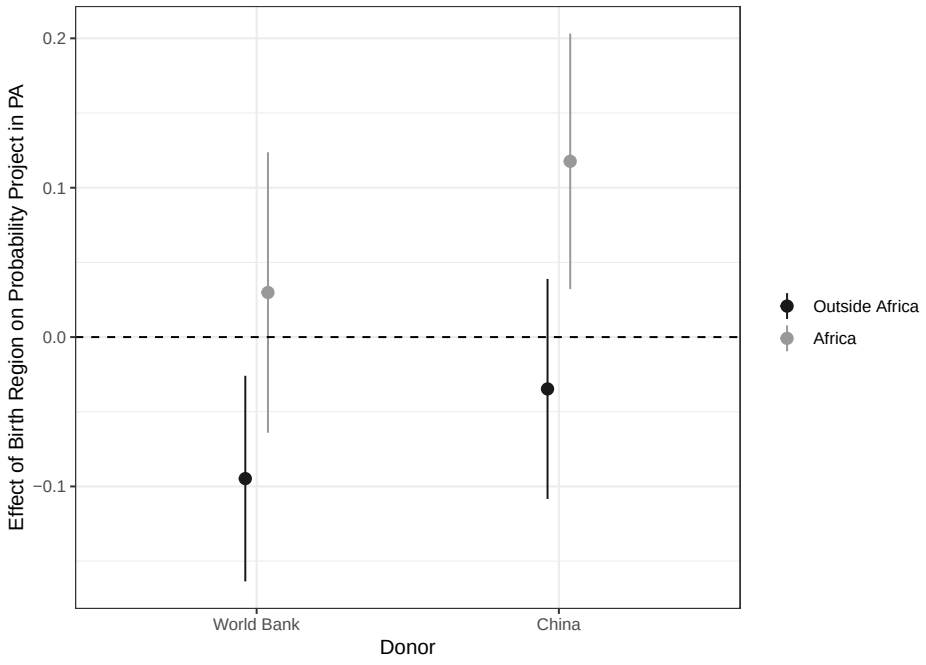
To account for these differences described in Table C.11 and to better understand whether the different results in Table 2 are due to institutional differences as we claim or just because of different

project characteristics, we pool the two samples and run the following model:

$$\begin{aligned}\text{ln PA}_{icst} = {} & \beta_1 \text{Birth Region}_{ct} + \beta_2 \text{China}_i + \beta_3(\text{Birth Region}_{ct} \times \text{China}_i) + \beta_4(\text{Birth Region}_{ct} \times \text{Africa}_s) \\ & + \beta_5(\text{China}_i \times \text{Africa}_s) + \beta_6(\text{Birth Region}_{ct} \times \text{China}_i \times \text{Africa}_s) \\ & + \delta \mathbf{X}_c' + \lambda \text{Sector}_i + \gamma \text{Num. Regions}_i + \zeta_{st} + \epsilon_{icst}\end{aligned}$$

The results in Figure C.5 are consistent with our main findings. After controlling for project characteristics, birth region shows no evidence of more selective enforcement among World Bank projects; if anything, the coefficient is negative and statistically significant outside of Africa. By contrast, within the African sample, we replicate the positive and significant birth-region effect among Chinese aid projects. Together, these results reinforce our interpretation that the divergence between World Bank and Chinese projects reflects institutional differences in how the two donors manage political favoritism.

Figure C.5: Birth Region Effect by Donor and Sample

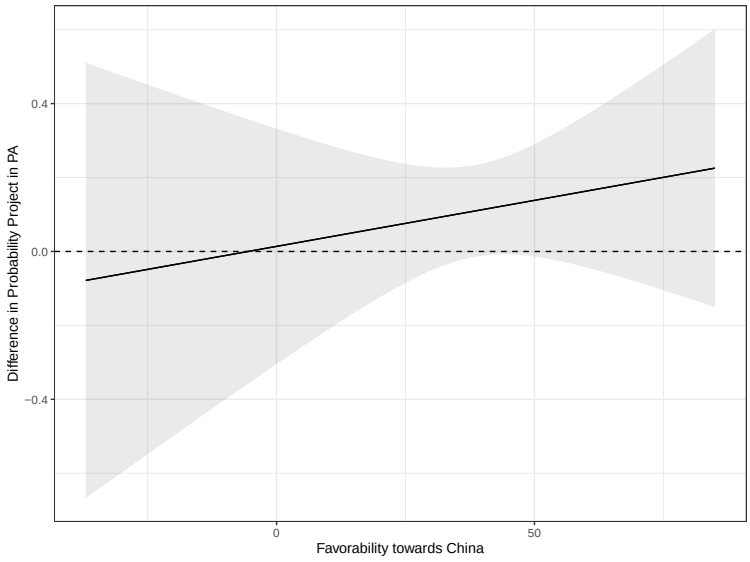


Note: Figure includes point estimates of birth region with 95%. Standard errors are clustered by recipient country. The models include the following covariates: Number of ADM1 regions covered by project, number of mines, number of oil and gas fields, log area, log population in 2000, log average night time light, and number of PAs. The models also include recipient-year and project sector fixed effects.

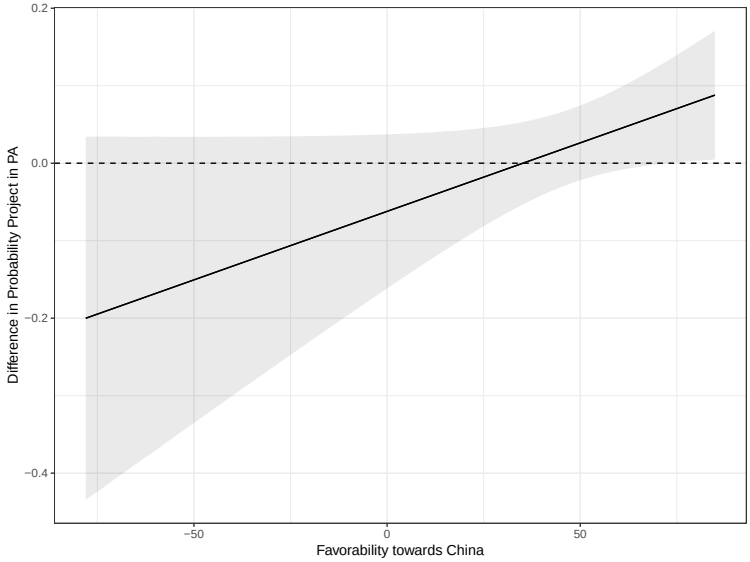
C.9 Favorability toward China as Moderator in Africa and Non-Africa Sample

Figure C.6: Favorability toward China as Moderator: Comparing Africa and Non-Africa

(a) Africa Sample



(b) Non-Africa Sample



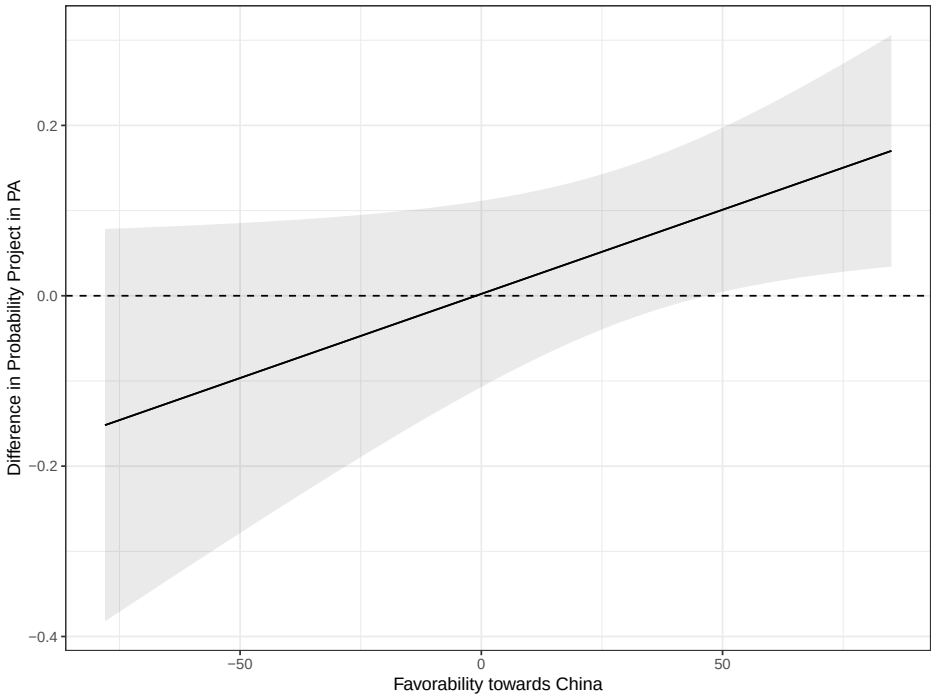
Note: Figure shows marginal effects of birth region on placement in PA conditional public opinion towards China in Africa sample (left panel) and non-Africa sample (right panel) with 95% confidence intervals. Standard errors are clustered by recipient country. The models also include recipient-year and project sector fixed effects.

Table C.12: Full Results Favorability toward China as Moderator

Dependent Variable:		In PA	
	All	Africa	No Africa
Model:	(i)	(ii)	(iii)
Variables			
Birth Region	-0.036	0.014	-0.062
	(0.045)	(0.162)	(0.051)
Num. ADM1 Regions	0.032	0.064**	0.018
	(0.018)	(0.019)	(0.012)
Num. Mines	0.004	0.004	0.004
	(0.003)	(0.004)	(0.003)
Num. Oil/Gas	0.012*	0.013	0.018**
	(0.005)	(0.013)	(0.005)
Log Population	0.042*	0.039**	0.051
	(0.018)	(0.013)	(0.037)
Log Total Area	0.000	-0.015	-0.028
	(0.019)	(0.024)	(0.034)
Num. PAs	0.001	0.000	0.001
	(0.001)	(0.001)	(0.001)
Log Avg. Nighttime Light	-0.103**	-0.131*	-0.124
	(0.037)	(0.052)	(0.075)
In Capital	0.086	0.046	0.079
	(0.050)	(0.059)	(0.061)
Birth Region × Net Favorability China	0.002*	0.002	0.002
	(0.001)	(0.004)	(0.001)
Fixed-effects			
Recipient-Year	Yes	Yes	Yes
Sector	Yes	Yes	Yes
Fit statistics			
R^2	0.596	0.591	0.643
Observations	2,024	1,013	1,011

Clustered (Recipient) standard-errors in parentheses
*Signif. Codes: **: 0.01, *: 0.05*

Figure C.7: Results with Non-Imputed Values for Favorability toward China

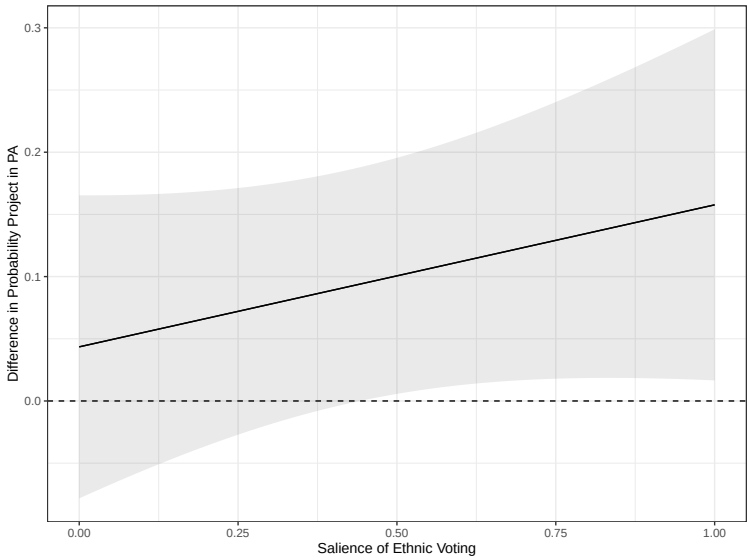


Note: Figure shows marginal effects of birth region on placement in PA conditional on public opinion towards China with 95% confidence intervals. Values for public opinion toward China are non-imputed, but value of last observation is projected forward for missing values. Standard errors are clustered by recipient country. The models also include recipient-year and project sector fixed effects.

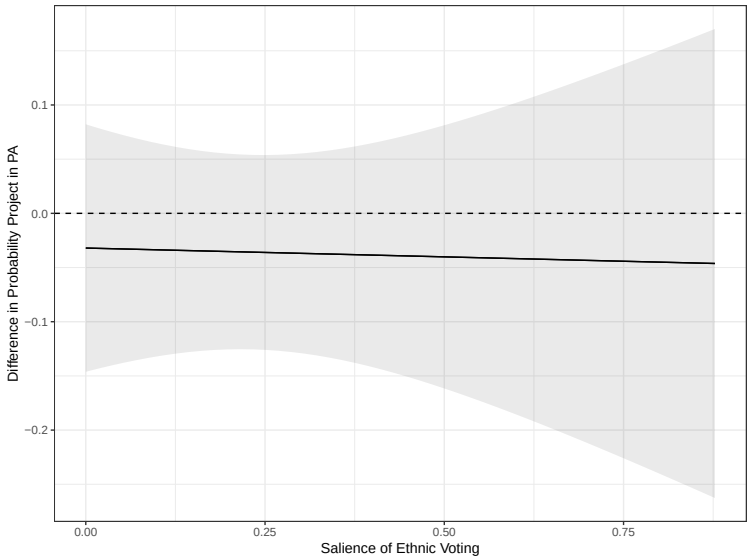
C.10 Ethnic Salience as Moderator in Africa and Non-Africa Sample

Figure C.8: Ethnic Salience as Moderator: Comparing Africa and Non-Africa

(a) Africa Sample



(b) Non-Africa Sample



Note: Figure shows marginal effects of birth region on placement in PA conditional on ethnic salience in Africa sample (left panel) and non-Africa sample (right panel) with 95% confidence intervals. Standard errors are clustered by recipient country. The models also include recipient-year and project sector fixed effects.

Table C.13: Full Results Ethnic Voting as Moderator

Dependent Variable:		In PA	
	All	Africa	No Africa
Model:	(i)	(ii)	(iii)
Variables			
Birth Region	-0.024	0.044	-0.032
	(0.041)	(0.062)	(0.058)
Num. ADM1 Regions	0.027*	0.063**	0.018
	(0.013)	(0.015)	(0.011)
Num. Mines	0.001	-0.003	0.012
	(0.003)	(0.002)	(0.007)
Num. Oil/Gas	0.009	-0.005	0.018**
	(0.005)	(0.014)	(0.006)
Log Population	0.025	0.046	0.022
	(0.025)	(0.038)	(0.034)
Log Total Area	0.033	0.018	0.021
	(0.023)	(0.032)	(0.039)
Num. PAs	0.001**	0.001	0.001
	(0.000)	(0.000)	(0.001)
Log Avg. Nighttime Light	-0.023	-0.027	-0.011
	(0.046)	(0.070)	(0.059)
In Capital	0.025	-0.050	0.044
	(0.042)	(0.050)	(0.058)
Birth Region × Ethnic Political Voting	0.192*	0.114	-0.016
	(0.082)	(0.093)	(0.155)
Fixed-effects			
Recipient-Year	Yes	Yes	Yes
Sector	Yes	Yes	Yes
Fit statistics			
R^2	0.610	0.623	0.636
Observations	2,025	964	1,061

Clustered (Recipient) standard-errors in parentheses
*Signif. Codes: **: 0.01, *: 0.05*

D Adjusting for Post-Treatment Bias

Assumptions

Studying the effect of race on use of force in police encounters, Knox, Lowe, and Mummolo (2020) describe a procedure for estimation of the average treatment effect while conditioning on a post-treatment moderator (in their case, whether an encounter occurred at all). We adapt this procedure to our study, allowing us to bound the ATE of a leader's birth region on the placement of a Chinese project in a PA (PA), conditional on a project being placed (which is itself a function of a leader's birth region). We formally describe the procedure in our context below.

Let the outcome Y_i be a dummy variable that takes value 1 if a project i is placed in a PA and 0 otherwise, the moderator M_i be a dummy that takes value 1 if a project is implemented and 0 otherwise, and D_i be a dummy that takes value 1 if the region is the birth region of a leader and 0 otherwise. We have four principal strata corresponding to four possible profiles of $M_i(d)$, which are described in Table D.1.[36] Definition of these principal strata allows to move to discussion of the four assumptions necessary for estimation of the $ATE_{M=1}$, or ATE of birth region among implemented projects.

	$M_i(0) = 0$	$M_i(0) = 1$
$M_i(1) = 0$	Never Implement	Implement if Not Birth Region
$M_i(1) = 1$	Implement if Birth Region	Always Implement

Table D.1: Mediator Value and Treatment Status

Assumption 1 (Project Observability) $Y_i(d, 0) = 0$ *for all* i *and for* $d \in \{0, 1\}$

This assumption states that unplaced (or unobserved) projects do not appear in PAs. Certainly, if we do not observe a project in our data, we will not observe that it is in a PA. This is thus centrally a question of data quality. It is possible that the data around Chinese projects is incomplete, which would violate this assumption. It is unlikely, however, that unobserved projects are likely to be the kinds of projects that end up in PAs. Projects in PAs are generally in infrastructure and other easily observable sectors, while unobserved projects should tend to be smaller and in less observable projects, such as consulting services for the government.

Assumption 2 (Mediator Monotonicity) $M_i(1) \geq M_i(0)$ *for all* i.

36. See Figure 2 in Knox et al.

This assumption rules out the existence of one of our principal strata, *Implement if Not Birth Region*. Intuitively, it is unlikely that projects are implemented if and only if the region is not the leader's birth region; any projects that are implemented in control regions would also be implemented in the treatment region. This is consistent both with the intuition of aid as a political tool and with existing empirical work on the subject (Dreher et al. 2019).

Assumption 3 (Differential Underlying Project Traits) $E[Y_i(d,m) \mid D_i = d', M_i(1) = M_i(0) = 1, X_i = x] \geq E[Y_i(d,m) \mid D_i = d', M_i(1) > M_i(0), X_i = x]$

Assumption 3 requires that projects approved only because they benefit a leader's home region are not systematically more likely to fall inside protected areas than projects that would have been implemented regardless. We cannot verify this directly, but several considerations make it plausible, and they turn on the type of project that owes its existence to political favoritism.

The favoritism-driven projects appear to be concentrated in sectors that are least likely to require encroachment on a protected area. Dolan and McDade (2020) show that a leader's birth region attracts disproportionately more Chinese aid in social sectors such as health and education, while finding no comparable birth-region effect in transport or communication. Social projects of this kind are typically smaller, population-proximate, and flexible in their siting. A clinic or school can be located to serve local residents rather than pushed through an ecologically sensitive area. So violating a protected area to build them would be an avoidable and largely gratuitous cost. Always-implemented projects, by contrast, are disproportionately the large, network-forming works whose routing is fixed by engineering and demand rather than politics, and which are therefore far more likely to intersect protected areas by construction.

This distinction is visible in our own data. Only 15% of social-sector projects overlap with a protected area, compared with 37.5% of communication and water-and-sanitation projects and 26.8% of transport projects. Sectors that show no sensitivity to a leader's birth region. The projects most prone to protected-area overlap are thus precisely those least associated with favoritism, which is the ordering Assumption 3 requires.

Finally, the pattern of our own results points the same way. The birth-region effect is concentrated in the strictest protected-area categories and the highest-stakes sectors, not in the loose categories and low-value sectors we would expect if the estimate were being driven by an influx of protected-area-prone favoritism projects.

Assumption 4 (Treatment Ignorability)

1. *With respect to potential mediator* $M_i(d) \perp\!\!\!\perp D_i \mid X_i$.

2. *With respect to potential outcomes:* $Y_i(d, m) \perp\!\!\!\perp D_i \mid M_i(0) = m', M_i(1) = m'', X_i$.

This represents a standard assumption of empirical work, indicating that both the treatment (birth region) is as good as randomly assigned to both the mediator (4.1) and the potential outcomes, conditional on some covariate(s) X_i. This assumption necessitates the inclusion of covariates that affect both the likelihood that a region produces a president and the likelihood that a region receives aid projects (and whether those projects are in PAs). We justify a number of controls and includes them in our analyses.

Results

Knox, Lowe, and Mummolo (2020) define the sharp bounds of the $ATE_{M=1}$ the following way.

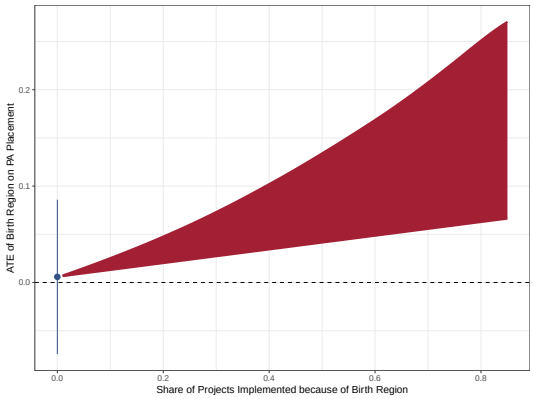
$$\mathbb{E}[\hat{\Delta}] + \rho\mathbb{E}[Y_i|D_1 = 0, M_i = 1](1 - \Pr(D_i = 0|M_i = 1)) \leq ATE_{M=1} \leq$$
$$\mathbb{E}[\hat{\Delta}] + \frac{\rho}{1-\rho}\left(\mathbb{E}[Y_i|D_1 = 1, M_i = 1] - \max\left\{0, 1 + \frac{1}{\rho}\mathbb{E}[Y_i|D_1 = 1, M_i = 1] - \frac{1}{\rho}\right\}\right) \times$$
$$\Pr(D_i = 0|M_i = 1) + \rho\mathbb{E}[Y_i|D_1 = 0, M_i = 1](1 - \Pr(D_i = 0|M_i = 1))$$

Note that $\hat{\Delta}$ refers to the "naive" estimator of the ATE under post-treatment bias defined as $\overline{Y_i|D_i = 1, M_i = 1} - \overline{Y_i|D_i = 0, M_i = 1}$. ρ refers to the share of projects that have only been implemented because they happen the leader's birth region.

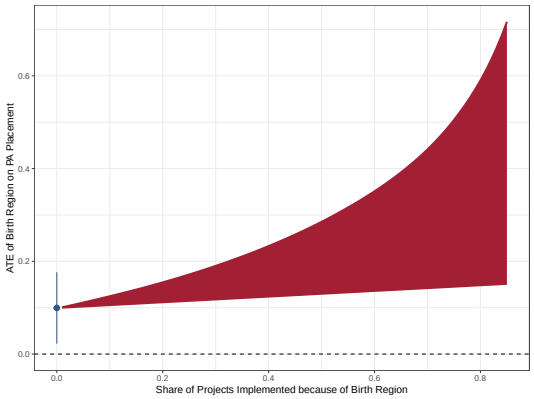
The ATE bounds for our projects are displayed in Figure D.1. The red shaded area represents the range of ATEs possible depending on the share of projects only happening because they are placed in the leader's birth region.

Figure D.1: ATE Bounds of birth region using Knox, Lowe, and Mummolo (2020) method

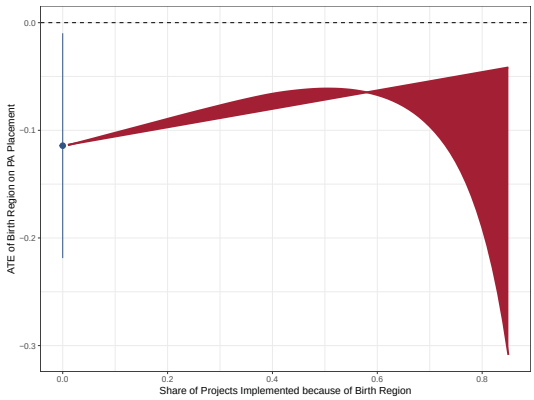
(a) All



(b) Africa



(c) Outside Africa



E Other Plausible Explanations

E.1 Birth Region to Non Birth Region Differences

Tables E.1 and E.2 document systematic differences between birth regions and non-birth regions. We compare basic geographic and economic characteristics of birth regions and non birth regions in two samples. In Table E.1, we look at how the characteristics differ between projects that benefit a birth region and projects that do not benefit any birth region. Table E.2 compares the differences across all geographic ADM1 regions independent of the project.

Table E.1: Balance table: Comparing Birth Region Projects to Non-Birth Region Projects

Variable	Mean Birth Region	Mean Non-Birth Region	Diff.-in-Means	t Stat.	$p < 0.05$
Number of PAs	27.90	16.98	10.93	-4.59	Yes
Commitment Value (IHS)	18.14	18.27	-0.13	1.36	No
Energy Proj.	0.18	0.19	-0.02	1.15	No
Capital	0.60	0.35	0.25	-13.21	Yes
Other Industry Proj.	0.04	0.05	-0.01	1.12	No
Log Area	9.82	9.44	0.38	-4.45	Yes
Log Night Time Lights	1.74	1.75	-0.00	0.20	No
Log Population	14.99	14.31	0.68	-10.61	Yes
Log Population Density	5.19	4.91	0.29	-4.34	Yes
Number of Mines	3.01	1.87	1.14	-3.58	Yes
Mining Proj.	0.07	0.13	-0.06	5.41	Yes
Number Oil Fields	1.70	0.83	0.88	-6.71	Yes
Comm. & Sanitation Proj.	0.12	0.08	0.05	-3.92	Yes
Social Proj.	0.38	0.33	0.05	-2.91	Yes
Transport Proj.	0.19	0.21	-0.02	1.06	No

Table E.2: Balance table: Comparing ADM1 Birth Regions to Non-Birth Regions

Variable	Mean Birth Region	Mean Non Birth Region	Difference-in-Means	t Statistic	$p < 0.05$
Number of PAs	11.65	10.31	1.34	-0.88	No
Capital	0.20	0.03	0.17	-7.60	Yes
Log Area	9.00	8.54	0.47	-3.71	Yes
Log Night Time Lights	1.55	1.32	0.23	-4.46	Yes
Log Population	13.77	12.78	0.99	-9.62	Yes
Log Population Density	4.77	4.25	0.52	-4.50	Yes
Number of Mines	1.20	0.78	0.42	-1.60	No
Number Oil Fields	0.45	0.53	-0.08	1.32	No

E.2 Governance Quality

A common concern is whether our main results vary with widely used governance indicators. In Table E.3, we interact the birth–region indicator with four V-Dem measures—democracy, corruption, clientelism, and rule of law. Across all specifications, none of these variables significantly moderates the birth–region effect: the interaction terms are uniformly statistically indistinguishable from zero.

Table E.3: Effect of Birth Region and Governance

Dependent Variable:	In PA			
Model:	(i)	(ii)	(iii)	(iv)
Variables				
Birth Region	-0.063	0.155	0.155	-0.059
	(0.100)	(0.094)	(0.120)	(0.080)
Num. ADM1 Regions	0.041**	0.041**	0.041**	0.041**
	(0.016)	(0.015)	(0.015)	(0.016)
Num. Mines	0.002	0.002	0.002	0.002
	(0.002)	(0.002)	(0.002)	(0.002)
Num. Oil/Gas	0.011*	0.011*	0.011*	0.011*
	(0.005)	(0.005)	(0.005)	(0.005)
Log Population	0.038*	0.039*	0.038*	0.038*
	(0.016)	(0.016)	(0.016)	(0.016)
Log Total Area	0.028	0.028	0.028	0.027
	(0.015)	(0.015)	(0.015)	(0.015)
Num. PAs	0.001*	0.001*	0.001*	0.001*
	(0.000)	(0.000)	(0.000)	(0.000)
Log Avg. Night Time Light	-0.048	-0.046	-0.046	-0.049
	(0.031)	(0.031)	(0.031)	(0.030)
In Capital	0.048	0.047	0.046	0.048
	(0.039)	(0.040)	(0.040)	(0.040)
Birth Region × Electoral Democracy	0.183			
	(0.187)			
Birth Region × Corruption Index		-0.212		
		(0.147)		
Birth Region × Clientelism Index			-0.245	
			(0.194)	
Birth Region × Rule of Law				0.195
				(0.153)
Fixed-effects				
Recipient-Year	Yes	Yes	Yes	Yes
Sector	Yes	Yes	Yes	Yes
Fit statistics				
R^2	0.580	0.581	0.581	0.581
Observations	3,660	3,660	3,660	3,660

Clustered (Recipient) standard-errors in parentheses
*Signif. Codes: **: 0.01, *: 0.05*

Appendix References

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